

# Trigonometric Solutions of the Yang-Baxter Equation, Nets, and Hypergeometric Functions

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*Dedicated to Israel Moiseevich Gelfand on his 80th birthday*

## 0. Introduction

It is now well known that the topology of knots in Euclidean 3-space is closely related to the study of the quantum Yang-Baxter equation

$$R_{12}R_{23}R_{12} = R_{23}R_{12}R_{23} \quad (0.a)$$

where  $R$  is an endomorphism of  $V \otimes V$  for a vector space  $V$ . The solutions of this equation, i.e.,  $R$ -matrices, give rise to topological invariants of knots. To compute the invariant determined by  $R$ , we present an oriented knot in  $\mathbb{R}^3$  by a diagram in the plane and assign  $R$  (resp.  $R^{-1}$ ) to every positive (resp. negative) crossing of the diagram. Contracting these matrices along the 1-strata of the diagram, we obtain a number. The Yang-Baxter equation guarantees that this number is preserved under the third Reidemeister move on knot diagrams. Under favorable conditions, this number is preserved under other Reidemeister moves and yields a topological invariant of knots. In this way we can obtain the Jones polynomial of knots and its various generalizations. This theory has a number of important ramifications, including a computation of the invariant in terms of  $6j$ -symbols. Moreover, this theory leads to invariants of 3-manifolds, 2-dimensional modular functors, and 3-dimensional topological quantum field theories; see [Tu] and references therein.

While the relationships between knots and equation (0.a) are understood quite well, it is by no means clear what is the topological counterpart of the Yang-Baxter equation with spectral parameter

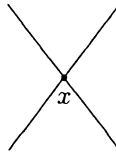
$$R_{12}(x) R_{23}(x+y) R_{12}(y) = R_{23}(y) R_{12}(x+y) R_{23}(x) \quad (0.b)$$

where  $x, y \in \mathbb{C}$ . The standard scheme producing knot invariants from the solutions of (0.a) does not apply to solutions of (0.b). Indeed, there is no natural and nontrivial way to assign the spectral parameter to a crossing of a knot diagram. Thus,  $R$ -matrices with spectral parameter do not yield new invariants of classical knots. This observation made at the early stages of

development of the theory of quantum knot invariants discouraged the study of  $R$ -matrices with spectral parameter in the topological context.

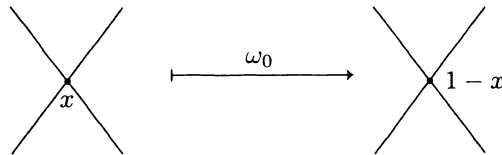
The aim of this paper is to revive the topological study of  $R$ -matrices with spectral parameter. We believe that this should eventually lead to a combinatorial definition of topological invariants of smooth 4-manifolds. However, a lot of work remains to be done in order to achieve this goal.

We introduce in this paper new geometric objects which are, in our opinion, topological counterparts of  $R$ -matrices with spectral parameter. We call these objects nets or plane nets. A simplest example of a net is a finite system of generic circles immersed in the plane such that, at each crossing, one of the four adjacent regions of the complement is distinguished and labelled with a number  $x \in \mathbb{C}$  as in Figure 0.1. We call  $x$  the formal angle of the crossing. We warn that the formal angle has nothing to do with the Euclidean angle between the tangent vectors of the circles in the crossing point.



**Figure 0.1**

Although there are no over-/undercrossings in the theory of nets, we can imitate knot theory and define analogues of Reidemeister moves for nets. There is one additional move which replaces the distinguished region at a crossing with the adjacent region (moving in the anticlockwise direction around the crossing) and replaces  $x$  with  $1 - x$  (see Figure 0.2).



**Figure 0.2**

As in the classical theory, we can consider colored nets, i.e., nets whose components are endowed with nonnegative integers, the “colors.” We also introduce more general relative nets and graph-nets which are analogues of tangles and knotted graphs in  $\mathbb{R}^3$ .

The standard scheme deriving knot invariants from  $R$ -matrices may be applied to nets using the  $R$ -matrices  $R(x)$  where the formal angle plays the role of the spectral parameter. We shall focus our attention on one specific  $R$ -matrix  $R(x) : V \otimes V \rightarrow V \otimes V$  which is the solution of (0.b) corresponding

to the quantum affine algebra  $U_q(\hat{sl}_2(\mathbb{C}))$  where  $q \in \mathbb{C} \setminus \{0\}$ . Here  $V$  is a 2-dimensional complex vector space with fixed basis. The isomorphism  $R(x)$  is given (with respect to the product basis) by the matrix

$$R(x) = \begin{bmatrix} [x-1] & 0 & 0 & 0 \\ 0 & -q^x & [x] & 0 \\ 0 & [x] & -q^{-x} & 0 \\ 0 & 0 & 0 & [x-1] \end{bmatrix} \quad (0.c)$$

with

$$[x] = \frac{q^x - q^{-x}}{q - q^{-1}} = \frac{\sin(\sqrt{-1}tx)}{\sin(\sqrt{-1}t)}$$

where we fix a logarithm  $t = \log(q) \in \mathbb{C}$  and set  $q^x = \exp(tx)$ .) This  $R$ -matrix has a topological interpretation similar to Kauffman's interpretation of the  $R$ -matrix without spectral parameter, see [KL]. The topological language will allow us to avoid lengthy algebraic computations.

We use the  $R$ -matrix  $R(x)$  (or rather its topological version) to define an invariant of nets. We call this invariant of a net  $L$  the bracket function of  $L$  and denote it  $\langle L \rangle$ . This invariant is a rational function of  $q$  and  $q^x$ , where  $x$  runs over the formal angles of the net. We show that under the Reidemeister moves on nets the bracket function is preserved up to multiplication by standard factors.

The next step is to extend the  $R$ -matrix (0.c) and the corresponding bracket function to nets colored with non-negative integers. This is done using a geometric version of the technique of fusion introduced in [KRS] for tensor powers of the 2-dimensional  $U_q(\hat{sl}_2(\mathbb{C}))$ -module  $V$ . Our version of fusion uses the Jones-Wenzl idempotents rather than standard symmetrizers acting in the tensor powers of  $V$ . Finally, we extend the bracket function to relative nets and graph-nets.

The study of the  $R$ -matrix (0.c) and its generalizations leads naturally to so-called trigonometric  $6j$ -symbols (with spectral parameter). They are enumerated by tuples of 6 nonnegative integers satisfying a few simple conditions. Each trigonometric  $6j$ -symbol is a rational function of  $\sin(\sqrt{-1}tx)$ ,  $\cos(\sqrt{-1}tx)$  with complex coefficients. (As above,  $x \in \mathbb{C}$  is the spectral parameter and  $t = \log(q) \in \mathbb{C}$ . Usually one fixes  $t$  and considers  $6j$ -symbols as functions of  $x$ .) The trigonometric  $6j$ -symbols first appeared as the Boltzman weights of the  $XXZ$ -models of statistical mechanics (for a survey, see [Ji]). They are closely related to more familiar  $6j$ -symbols without spectral parameter (see [KR]) which may be obtained as limits of the trigonometric  $6j$ -symbols when  $x \mapsto \pm\infty$ .

We introduce here a topological approach to the trigonometric  $6j$ -symbols. This allows us to rederive their fundamental properties including the Yang-

Baxter identity and the orthogonality. We also establish their full tetrahedral symmetry. This result is new and may have interesting applications in statistical mechanics. A related natural problem is to establish, for the trigonometric  $6j$ -symbols, an analogue of the pentagon identity for the  $6j$ -symbols without spectral parameter.

The trigonometric  $6j$ -symbols play in the theory of nets the same role as the  $6j$ -symbols without spectral parameter in the theory of knots. We discuss a face state sum model based on the trigonometric  $6j$ -symbols and compute the bracket function of a net.

Finally, we identify the trigonometric  $6j$ -symbols with certain basic hypergeometric functions. It is known that the  $6j$ -symbols without spectral parameter can be expressed as balanced basic hypergeometric series  ${}_4\phi_3$ , see [KR]. By [GR], they can also be rewritten as a very-well-poised series  ${}_8\phi_7$ . We show that up to multiplication by a monomial, a trigonometric  $6j$ -symbol is a balanced very-well-poised series  ${}_{10}\phi_9$ . The tetrahedral symmetry of the trigonometric  $6j$ -symbols implies one of the most important symmetries in the theory of hypergeometric functions, known as the Bailey transform. A computation of the trigonometric  $6j$ -symbols in a special case implies the Jackson summation formula containing most of the known summation formulas in the theory of the hypergeometric functions.

It was discovered by R. Baxter that the matrix  $R(x)$  in (0.c) admits an elliptic deformation (see [Ba]). He also reformulated this deformation in terms of elliptic  $6j$ -symbols interpreted as the Boltzman weights of the  $XYZ$ -models of statistical mechanics. More general elliptic  $6j$ -symbols were introduced in [DJMO]. In the subsequent paper [FT], we shall generalize the fundamental identities for the trigonometric  $6j$ -symbols (including the tetrahedral symmetry) to the elliptic case. The relations to the basic hypergeometric series established in this paper leads to a definition of a new class of “elliptic” hypergeometric series singled out by their modular properties. The identities for elliptic  $6j$ -symbols imply corresponding identities for these series.

The paper consists of 9 sections. In Section 1 we introduce nets. In Section 2 we define and study their bracket functions. In Section 3 we briefly discuss the Jones-Wenzl idempotents in the Temperley-Lieb algebras; they are used in Section 4 to define the bracket function for colored nets. In Section 4 we also state a theorem describing the behavior of the bracket under basic moves on nets; this theorem is proven in Section 5. In Section 6 we introduce the trigonometric  $6j$ -symbols and study their fundamental properties: symmetries, the Yang-Baxter identity, and the orthonormality relation. In Section 6 we also discuss a face model for the bracket function of a net. In Section 7 we establish the tetrahedral symmetry of trigonometric  $6j$ -symbols. In Section 8 we give explicit formulas computing these  $6j$ -symbols. Finally, in Section 9 we identify the trigonometric  $6j$ -symbols with basic hypergeometric series and

relate the symmetries of  $6j$ -symbols with the transformation properties of the hypergeometric series.

We would like to acknowledge the hospitality and financial support of the Newton Mathematical Institute, Louis Pasteur University, and Yale University where parts of this work were done. The first author was partially supported by the NSF grant DMS-9400908.

We are happy to dedicate this paper to Israel Moiseevich Gelfand whose work influenced considerably our understanding of mathematics.

## 1. Plane nets

### 1.1. Definition of nets

A net in  $\mathbb{R}^2$  is a finite system of circles immersed in  $\mathbb{R}^2$  with only double transversal crossings such that at each crossing one of the four adjacent regions of the complement is distinguished and labelled with a complex number as in Figure 0.1.

Two nets in  $\mathbb{R}^2$  are isotopic if they may be obtained from each other by an ambient isotopy of the plane.

### 1.2. Moves on nets

In Figure 1.1 we describe three moves on nets which are the analogues of the Reidemeister moves on knot diagrams. It is understood that these moves apply to a part of a net lying in a small disk and keep the remaining part of the net unchanged. The third move applies only when  $z = x + y$ . The moves in Figures 0.2 and 1.1 are denoted  $\omega_0, \omega_1, \omega_2, \omega_3$ , respectively. The inverse moves are denoted  $\omega_0^{-1}, \omega_1^{-1}, \omega_2^{-1}, \omega_3^{-1}$ .

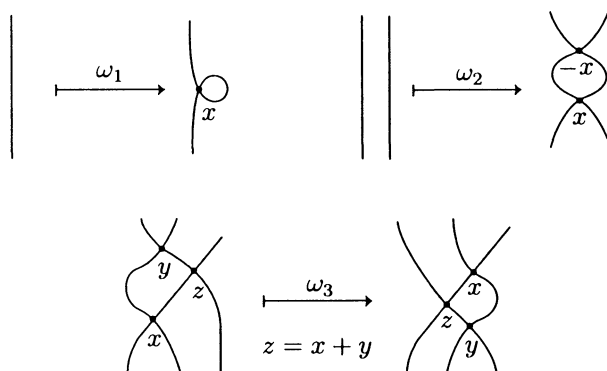


Figure 1.1

In principle, we are interested in the invariants of nets preserved under the moves  $\{\omega_i \mid i = 0, 1, 2, 3\}$ . However, one should not push the analogy with knot

theory too far. Knot diagrams and Reidemeister moves are just convenient tools to describe fundamental geometric objects — knots in Euclidean 3-space. A similar global interpretation of nets is not available. Therefore we do not insist on following knot theory too closely and, in particular, shall be satisfied with invariants of nets which change under the basic moves in a reasonable way.

### 1.3. Relative nets

Relative nets generalize nets: besides circles we admit arcs whose endpoints lie on the horizontal lines  $\mathbb{R} \times 0$  and  $\mathbb{R} \times 1$ . A relative net with  $k \geq 0$  inputs and  $l \geq 0$  outputs (or, briefly, a  $(k, l)$ -net) consists of a finite number of circles and  $(k+l)/2$  arcs immersed in  $\mathbb{R} \times [0, 1]$  such that (i) all crossings are double and transversal and at each crossing one of the four incident regions of the complement is distinguished and labelled with a complex number as in Figure 0.1; (ii) the endpoints of the arcs are  $(1, 0), \dots, (k, 0), (1, 1), \dots, (l, 1)$ ; (iii) the circles do not meet the lines  $\mathbb{R} \times 0, \mathbb{R} \times 1$  and the arcs meet these lines orthogonally in the endpoints. It is clear that  $(0, 0)$ -nets are just nets as defined in Section 1.1.

Two relative nets are isotopic if they can be obtained from each other by an ambient isotopy of the strip  $\mathbb{R} \times [0, 1]$  constant on the boundary. The moves  $\omega_i$ ,  $i = 0, 1, 2, 3$ , on relative nets are defined in the same way as in Section 1.2, they proceed far from the boundary lines.

Similarly to the theory of tangles, relative nets form a category whose objects are nonnegative integers and whose morphisms are isotopy classes of nets (cf. [Tu, Chapter XII]). The composition of morphisms  $k \rightarrow l$  and  $l \rightarrow m$  is obtained by placing the corresponding  $(l, m)$ -net on the top of the  $(k, l)$ -net, identifying the common  $l$  endpoints, and compressing the result in the strip  $\mathbb{R} \times [0, 1]$ . Placing a relative net to the left of another one, we provide this category with a strictly associative tensor product. For objects, this tensor product is nothing but addition of integer numbers.

**1.4. Remark.** Instead of the rule  $x \mapsto 1 - x$  in the definition of the move  $\omega_0$  we can use the rule  $x \mapsto \text{const} - x$  with any fixed constant  $\text{const} \neq 0$ . Dividing all formal angles by  $\text{const}$  we reduce ourselves to the case  $\text{const} = 1$ .

## 2. Bracket function for nets

### 2.1. Bracket function

For a net  $L \subset \mathbb{R}^2$ , we consider a function  $\langle L \rangle = \langle L \rangle(q)$  of one variable  $q \in \mathbb{C} \setminus \{0\}$ . It is called a bracket function of  $L$  if it satisfies the following three conditions:

- (i) if  $L = \emptyset$  then  $\langle L \rangle = 1$ ;
- (ii) if  $\langle L \rangle$  is a disjoint union of a net  $L'$  and an embedded circle then  $\langle L \rangle = (-q - q^{-1})\langle L' \rangle$ ;

(iii) for any crossing of  $L$ , we have the equality in Figure 2.1.

These conditions need a few comments. The factor  $-q - q^{-1}$  plays the same role as the factor  $-A^2 - A^{-2}$  in Kauffman's model for the bracket polynomial of a framed link (which is a version of the Jones polynomial of an oriented link). In Figure 2.1 we smooth the crossing of  $L$  in two different ways, the one joining two regions adjacent to the distinguished region and the other one joining the distinguished region with the opposite region. We keep the remaining part of  $L$  and the formal angles at all other crossing points of  $L$ . This gives two new nets,  $L', L''$ , respectively. The equality in Figure 2.1 means that

$$\langle L \rangle = [x]\langle L' \rangle + [x - 1]\langle L'' \rangle. \quad (2.1.a)$$

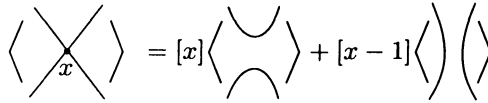


Figure 2.1

**2.2. Theorem.** *There exists a unique bracket function for nets. This function is invariant under (ambient) isotopies of nets.*

This theorem is essentially obvious: applying (2.1.a) we get rid of crossings and then applying (ii) inductively we reduce the computation to the case of an empty net.

The next theorem describes the behavior of the bracket function under the Reidemeister moves.

**2.3. Theorem.** *Let  $L_i$  be a net obtained from a net  $L$  by the transformation  $\omega_i$  with  $i = 0, 1, 2, 3$ . We have*

$$\begin{aligned} \langle L_0 \rangle &= -\langle L \rangle, \quad \langle L_1 \rangle = -[x - 2]\langle L \rangle, \\ \langle L_2 \rangle &= -[x + 1][x - 1]\langle L \rangle, \quad \langle L_3 \rangle = \langle L \rangle. \end{aligned}$$

This theorem will be proven (in a generalized form) in Section 2.5.

## 2.4. The Temperley-Lieb category

In order to extend the bracket function to relative nets, we need the so-called Temperley-Lieb category  $TL$ . For details and further references see [KL], [Tu, Chapter XII].

The objects of the category  $TL$  are nonnegative integers. For nonnegative integers  $k, l$ , the set of morphisms  $TL_{k,l} = \text{Hom}_{TL}(k, l)$  is a finite-dimensional vector space defined as follows. If  $k + l$  is odd then  $TL_{k,l} = 0$ . If  $k + l$  is even then we consider systems of disjoint  $(k + l)/2$  arcs embedded in  $\mathbb{R} \times [0, 1]$  with the endpoints  $(1, 0), \dots, (k, 0), (1, 1), \dots, (l, 1)$ . It is understood that the arcs meet the lines  $\mathbb{R} \times 0, \mathbb{R} \times 1$  orthogonally in the endpoints. Up to ambient isotopy in  $\mathbb{R} \times [0, 1]$  (constant on the boundary) there is only a finite number of such systems. These isotopy classes form a basis of  $TL_{k,l}$ . Thus,  $TL_{k,l}$  consists of formal linear combinations of systems of arcs as above with complex coefficients. For instance,  $\text{Hom}(0, 0) = \mathbb{C}$  is generated by the empty set of arcs.

The composition in the category  $TL$  depends on the choice of a non-zero  $q \in \mathbb{C}$ . For non-negative integers  $k, l, m$ , the composition  $TL_{k,l} \times TL_{l,m} \rightarrow TL_{k,m}$  is a bilinear pairing defined on the generators  $\alpha \in TL_{k,l}, \beta \in TL_{l,m}$  as follows. It suffices to consider the case where the numbers  $k + l$  and  $l + m$  are even. Placing the system of arcs  $\beta$  on the top of  $\alpha$  and compressing the result in the strip  $\mathbb{R} \times [0, 1]$  we obtain a 1-dimensional submanifold of the strip with the endpoints  $(1, 0), \dots, (k, 0), (1, 1), \dots, (m, 1)$ . This submanifold consists of  $(k + m)/2$  arcs and a certain number,  $N$ , of circles. The arcs represent a generator  $\gamma \in TL_{k,m}$ . We set  $\beta \circ \alpha = (-q - q^{-1})^N \gamma \in TL_{k,m}$ . It is easy to check that this is an associative composition law in  $TL$ . Placing a system of arcs to the left of another one, we provide  $TL$  with a strictly associative tensor product. For objects, this tensor product is addition of numbers (cf. Section 1.3).

The composition defined above endows the vector space  $TL_k = TL_{k,k}$  with the structure of an associative algebra over  $\mathbb{C}$ , the Temperley-Lieb algebra corresponding to  $q$ . It is multiplicatively generated by the unit element  $1_k$  represented by  $k$  vertical arcs and the elements  $\{e_i \mid i = 1, \dots, k - 1\}$  shown in Figure 2.2. Note that  $TL_k = \mathbb{C} \cdot 1_k \oplus \mathbb{C}(e_1, \dots, e_{k-1})$  where  $\mathbb{C}(e_1, \dots, e_{k-1})$  is the two-sided ideal of  $TL_k$  generated by  $e_1, \dots, e_{k-1}$ .

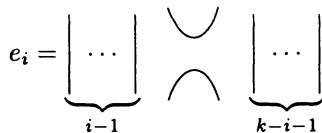


Figure 2.2



## 2.5. Bracket function for relative nets

We can extend the bracket function to relative nets. For a  $(k, l)$ -net  $L$ , its bracket function  $\langle L \rangle = \langle L \rangle(q)$  takes values in the vector space  $TL_{k,l} = \text{Hom}_{TL}(k, l)$ . It satisfies conditions (ii), (iii) of Section 2.1 and the following condition:

(i)' if  $L$  consists of disjoint embedded arcs, then  $\langle L \rangle = L \in TL_{k,l}$ .

In generalizing Theorem 2.2 we observe that there exists a unique bracket function for relative nets. It is invariant under (ambient) isotopies of nets. The bracket yields a covariant functor from the category of nets in the Temperley-Lieb category. This functor is the identity on the objects and preserves the tensor product.

The next theorem generalizes Theorem 2.3 to relative nets.

**2.6. Theorem.** *Let  $L_i$  be a relative net obtained from a relative net  $L$  by the transformation  $\omega_i$  with  $i = 0, 1, 2, 3$ . We have*

$$\langle L_0 \rangle = -\langle L \rangle, \quad (2.6.a)$$

$$\langle L_1 \rangle = -[x - 2]\langle L \rangle, \quad (2.6.b)$$

$$\langle L_2 \rangle = -[x + 1][x - 1]\langle L \rangle, \quad (2.6.c)$$

$$\langle L_3 \rangle = \langle L \rangle. \quad (2.6.d)$$

**Proof.** Equality (2.6.a) follows directly from definitions and the identity  $[-x] = -[x]$ . To prove (2.6.b) it is enough to consider the case where  $L$  consists of one vertical interval and  $L_1$  is obtained from  $L$  by  $\omega_1$ , as in Figure 1.1. We have

$$\langle L_1 \rangle = [x]\langle L' \rangle + [x - 1]\langle L'' \rangle$$

where  $L' = L$  and  $L''$  is the disjoint union of  $L$  and an embedded circle. Therefore  $\langle L' \rangle = L$ ,  $\langle L'' \rangle = -(q + q^{-1})L$ , and

$$\langle L_1 \rangle = ([x] - (q + q^{-1})[x - 1])L = -[x - 2]L.$$

The proof of (2.6.c) is obtained by an application of the recursive formula (2.1.a) twice. Let us prove (2.6.d). It is enough to consider the case where  $L, L_3$  are the relative  $(3, 3)$ -nets drawn in Figure 1.1 on the left and on the right of the arrow  $\omega_3$ . Recall the generators  $e_1, e_2$  of the Temperley-Lieb algebra  $TL_3$  with unit  $1_3$ . Note that  $e_i^2 = -(q + q^{-1})e_i = [-2]e_i$  for  $i = 1, 2$ . It follows from

definitions that

$$\begin{aligned}\langle L \rangle &= ([y]e_1 + [y-1]1_3) ([z]e_2 + [z-1]1_3) ([x]e_1 + [x-1]1_3) \\ &= [y][z][x]e_1 + [y][z][x-1]e_1e_2 + [y][z-1][x][-2]e_1 + [y][z-1][x-1]e_1 \\ &\quad + [y-1][z][x]e_2e_1 + [y-1][z][x-1]e_2 + [y-1][z-1][x]e_1 \\ &\quad + [y-1][z-1][x-1]1_3.\end{aligned}$$

Similarly,

$$\begin{aligned}\langle L_3 \rangle &= ([x]e_2 + [x-1]1_3) ([z]e_1 + [z-1]1_3) ([y]e_2 + [y-1]1_3) \\ &= [x][z][y]e_2 + [x][z][y-1]e_2e_1 + [x][z-1][y][-2]e_2 + [x][z-1][y-1]e_2 \\ &\quad + [x-1][z][y]e_1e_2 + [x-1][z][y-1]e_1 + [x-1][z-1][y]e_2 \\ &\quad + [x-1][z-1][y-1]1_3.\end{aligned}$$

Therefore  $\langle L \rangle - \langle L_3 \rangle = a(e_1 - e_2)$  where

$$a = [x][z][y] + [x-1][z-1][y] + [x][z-1][y-1] - [2][x][z-1][y] - [x-1][z][y-1].$$

It remains to show that  $a = 0$ . This follows from the assumption  $z = x + y$  and the identities

$$\begin{aligned}[x][y] - [x-1][y-1] &= [x+y-1], \\ [x+y] + [x-1][y] + [x][y-1] - [2][x][y] &= 0.\end{aligned}$$

## 2.7. Extension to formal linear combinations of nets

It will be convenient to consider formal linear combinations of  $(k, l)$ -nets with complex coefficients. The bracket function extends to them by linearity and takes values in  $TL_{k,l}$ .

**2.8. Remark.** There is a simple renormalization of the bracket function of a net  $L$  which makes it invariant with respect to the move  $\omega_0$ . To this end we define the degree of a point with respect to  $L$ . For  $X \in \mathbb{R}^2 \setminus L$ , the degree  $\deg_L(X) \in \mathbb{Z}/2\mathbb{Z}$  is the number (mod 2) of intersections of  $L$  with a generic path leading from  $X$  to infinity. This degree does not depend on the choice of the path and depends only on the region (connected component) of  $\mathbb{R}^2 \setminus L$  containing  $X$ . Therefore we can speak of the degree of such a region. It is clear that the degree of the only unbounded region of  $\mathbb{R}^2 \setminus L$  equals 0 and that two regions adjacent to a segment of  $L$  from different sides have different degrees. Let  $\deg(L)$  be the sum (mod 2) of the degrees of distinguished regions corresponding to all crossings of  $L$ . It follows from (2.6.a) that the function

$(-1)^{\deg(L)}\langle L \rangle$  is invariant under the move  $\omega_0$ . These definitions generalize to relative nets, although to define  $\deg$  we should choose everywhere either the “left” infinity  $-\infty \times [0, 1]$  of  $\mathbb{R} \times [0, 1]$  or the “right” infinity  $+\infty \times [0, 1]$ . Note that this renormalized bracket is invariant under  $\omega_3$  only up to sign.

### 3. The Jones-Wenzl idempotents re-examined

#### 3.1. The Jones-Wenzl idempotents

In the sequel we shall need an element  $f_k$  of the Temperley-Lieb algebra  $TL_k$  introduced by V. Jones [Jo] and further studied by H. Wenzl [We]. It is defined under the assumption that  $q^{2n} \neq 1$  for  $n = 1, 2, \dots, k$ . Then there exists a unique  $f_k \in TL_k$  such that  $f_k - 1_k \in \mathbb{C}(e_1, \dots, e_{k-1})$  and  $e_i f_k = f_k e_i = 0$  for all  $i = 1, \dots, k-1$  (for a proof, see e.g. [Tu, Chapter XII]). It follows from these conditions that  $f_k$  is an idempotent:

$$f_k^2 - f_k = (f_k - 1)f_k = 0. \quad (3.1.a)$$

For instance,  $f_1 = 1$ ,  $f_2 = 1_2 + (q + q^{-1})^{-1}e_1$ .

The idempotent  $f_k$  belongs to the center of  $TL_k$  and annihilates all additive generators of  $TL_k$  (represented by systems of arcs) except  $1_k$ . Indeed, all these generators are monomials in  $e_1, \dots, e_{k-1}$ .

We need to interpret the idempotent  $f_k$  in terms of the bracket function of nets. To this end we define a  $(k, k)$ -net  $\tilde{F}_k$  as follows. It consists of  $k$  embedded arcs in  $\mathbb{R} \times [0, 1]$  where the  $i$ -th arcs connects the points  $(i, 0)$  and  $(k - i + 1, 1)$ . We assume that the projection  $\mathbb{R} \times [0, 1] \mapsto [0, 1]$  maps each of these arcs homeomorphically onto  $[0, 1]$  and that the arcs have no triple intersections. By combinatorial reasons, every two arcs of this family meet at least once. We assume that every pair of arcs has exactly one transversal intersection. For  $i < j$ , we provide the crossing point of the  $i$ -th and  $j$ -th arcs with the distinguished region which lies below this point as in Figure 0.1 and we label this region with the (negative) number  $i - j$ . This gives the  $(k, k)$ -net  $\tilde{F}_k$ . Note that this net is defined only up to transformations  $\omega_3, \omega_3^{-1}$ . This is sufficient to consider the bracket  $\langle \tilde{F}_k \rangle \in TL_k$ . For instance,  $\tilde{F}_2$  is the net in Figure 0.1 with  $x = -1$ . Up to isotopy there are two nets representing  $\tilde{F}_3$ , they appear on the left and the right of the  $\omega_3$ -arrow in Figure 1.1 where  $x = y = -1, z = -2$ . A recursive formula for  $\tilde{F}_k$  is given in Figure 3.1.

#### 3.2. Lemma.

$$\langle \tilde{F}_k \rangle = \left( \prod_{1 \leq i < j \leq k} [i - j - 1] \right) f_k \in TL_k.$$

This Lemma implies that  $f_k$  belongs to the image of the bracket function

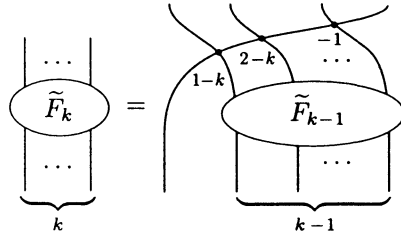


Figure 3.1

extended to formal linear combinations of  $(k, k)$ -nets. Indeed, if  $f_k$  is defined then  $\prod_{1 \leq i < j \leq k} [i - j - 1] \neq 0$  and therefore

$$f_k = \left( \prod_{1 \leq i < j \leq k} [i - j - 1] \right)^{-1} \langle \tilde{F}_k \rangle = \left\langle \prod_{1 \leq i < j \leq k} [i - j - 1]^{-1} \tilde{F}_k \right\rangle.$$

We shall use the notation

$$F_k = \left( \prod_{1 \leq i < j \leq k} [i - j - 1] \right)^{-1} \tilde{F}_k.$$

By abuse of language, we call  $F_k$  a net representing  $f_k$ .

**Proof of Lemma.** Set  $f = \langle \tilde{F}_k \rangle \in TL_k$ . Let us show that  $f e_i = e_i f = 0$  for any  $i = 1, \dots, k-1$ . Applying if necessary  $\omega_3, \omega_3^{-1}$  to  $\tilde{F}_k$  we may assume that  $\tilde{F}_k$  is the composition of two  $(k, k)$ -nets  $F' F''$  where  $F'' = 1_{i-1} \otimes \tilde{F}_2 \otimes 1_{k-i-1}$  is obtained from the simple crossing (as in Figure 0.1) with  $a = -1$  by adding  $i-1$  vertical intervals on the left and  $k-i-1$  vertical intervals on the right. By definition,  $\langle F'' \rangle = -e_i - [2]1_k \in TL_k$ . We have  $(-e_i - [2]1_k)e_i = -(e_i)^2 - [2]e_i = 0$ . Hence  $f e_i = \langle \tilde{F}_k \rangle e_i = \langle F' \rangle \langle F'' \rangle e_i = 0$ . Similarly,  $e_i f = 0$  for any  $i = 1, \dots, k-1$ . This implies that  $f \in \mathbb{C} f_k$ . To prove the lemma it remains to show that

$$f - \prod_{1 \leq i < j \leq k} [i - j - 1] \cdot 1_k \in \mathbb{C}(e_1, \dots, e_{k-1}). \quad (3.2.a)$$

Analyzing the definition of the bracket function for the net  $\tilde{F}_k$  drawn in Figure 3.1 it is easy to observe that there is only one smoothing of the crossings of  $\tilde{F}_k$  which contributes to  $f = \langle \tilde{F}_k \rangle$  a summand proportional to  $1_k$ . This is the vertical smoothing joining at each crossing the corresponding distinguished region with the opposite region. All other smoothings contribute summands belonging to  $\mathbb{C}(e_1, \dots, e_{k-1})$ . This directly implies (3.2.a) and the lemma.

#### 4. Bracket function for colored graph-nets

Fix a complex number  $q \neq 0$  which is not a root of unity.

##### 4.1. Colored nets

A (relative) net is said to be colored if each its underlying circle or arc is provided with a nonnegative integer. Note that in the representation-theoretic interpretation, an integer color  $m$  represents the irreducible  $U_q(\hat{\mathfrak{sl}}_2(\mathbb{C}))$ -module of complex dimension  $m + 1$ .

We define a cabling of a colored  $(k, l)$ -net  $L$  which transforms  $L$  into an uncolored net (or rather into a formal linear combination of uncolored nets). Observe that the crossings of  $L$  split  $L$  into a finite number of 1-strata which are disjoint embedded intervals or embedded circles. Each of these 1-strata lies on an arc or a circle of  $L$  and inherits its color. We replace every 1-stratum of color  $m$  with  $m$  parallel 1-strata and insert in this bunch an  $(m, m)$ -net (or a formal linear combination of  $(m, m)$ -nets) whose bracket is equal to  $f_m$ . For instance, we may use  $F_m$ , see Figure 4.1. Each crossing point of  $L$  is transformed as in Figure 4.2 where the  $mn$  crossing points are numerated by pairs  $(a, b)$  with  $a = 1, \dots, m; b = 1, \dots, n$  and the distinguished region adjacent to the  $(a, b)$ -th crossing point is labelled with

$$x_{a,b} = x + \frac{n-m}{2} + a - b. \quad (4.1.a)$$

This yields a formal linear combination of uncolored nets with complex coefficients. It is denoted by  $L^c$ .

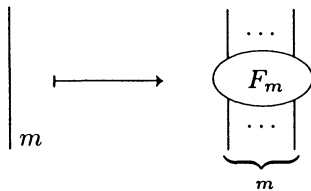


Figure 4.1

The net  $L^c$  is not uniquely defined because of the choice of nets representing the Jones-Wenzl idempotents. However the bracket function of  $L^c$  does not depend on this choice. Set

$$\langle L \rangle(q) = \langle L^c \rangle(q).$$

If  $k = l = 0$  then  $\langle L \rangle(q) \in \mathbb{C}$ . For general  $k, l$ , the bracket  $\langle L \rangle(q)$  is an element of the Temperley-Lieb module  $TL_{k', l'}$  where  $k'$  (resp.  $l'$ ) is the sum of colors of the inputs (resp. outputs) of  $L$ . It is obvious that  $\langle L \rangle(q)$  is invariant under

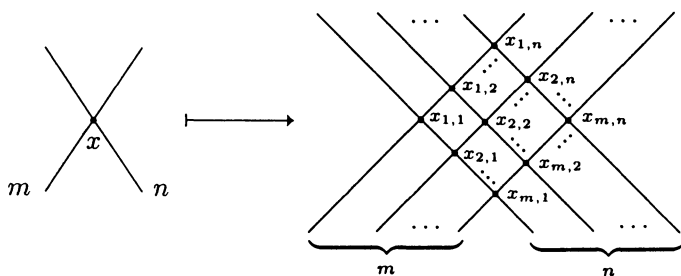


Figure 4.2

(ambient) isotopy of  $L$ . The bracket function yields a covariant functor from the category of colored relative nets in the Temperley-Lieb category.

#### 4.2. Extension to graph-nets

Graph-nets generalize nets and relative nets. We shall restrict ourselves to (finite) graphs whose vertices have valency  $\leq 3$ . Let  $k, l$  be nonnegative integers and  $g$  be a graph with  $k + l$  vertices of valency 1 and other vertices of valency 2 and 3. A graph-net based on  $g$  with  $k \geq 0$  inputs and  $l \geq 0$  outputs is an immersion  $g \rightarrow \mathbb{R} \times [0, 1]$  such that (i) all self-crossings of the immersion are double transversal intersections of edges such that at each crossing one of the four incident regions of the complement is distinguished and labelled with a complex number as in Figure 0.1; (ii) small neighborhoods of the vertices of  $g$  are embedded in  $\mathbb{R} \times [0, 1]$  and their images do not overlap; (iii) the vertices of valency 1 are mapped into the points  $(1, 0), \dots, (k, 0), (1, 1), \dots, (l, 1)$ ; (iv) the image of  $g$  meets the lines  $\mathbb{R} \times 0, \mathbb{R} \times 1$  orthogonally along these  $k + l$  points. Note that no formal angles are associated to vertices of  $g$ .

Isotopy and four basic moves on graph-nets are defined in the same way as in Section 1; they proceed far from the vertices of the graph.

A coloring of a graph-net assigns a nonnegative integer to each its edge such that (i) the colors of edges incident to a vertex of valency 2 are equal and (ii) if  $i, j, k$  are the colors of edges incident to a 3-valent vertex, then the triple  $(i, j, k)$  is admissible in the sense that  $|i - j| \leq k \leq i + j$  and  $i + j + k \equiv 0 \pmod{2}$ .

The cabling of colored nets extends to graph-nets as follows. The crossings and the trivalent vertices of a graph-net  $L$  split  $L$  into a finite number of 1-strata which are disjoint embedded intervals or embedded circles. Each of these 1-strata and the crossings of  $L$  are transformed in the same way as in Section 4.1. Each trivalent vertex incident to edges colored with  $i, j, k$  is transformed in the union of  $(i + j + k)/2$  disjoint intervals as in Figure 4.3, where  $r = (i + j - k)/2, s = (i + k - j)/2, t = (j + k - i)/2$ . In this way  $L$  is transformed into a formal linear combination of uncolored nets  $L^c$ . Set  $\langle L \rangle(q) = \langle L^c \rangle(q)$ .

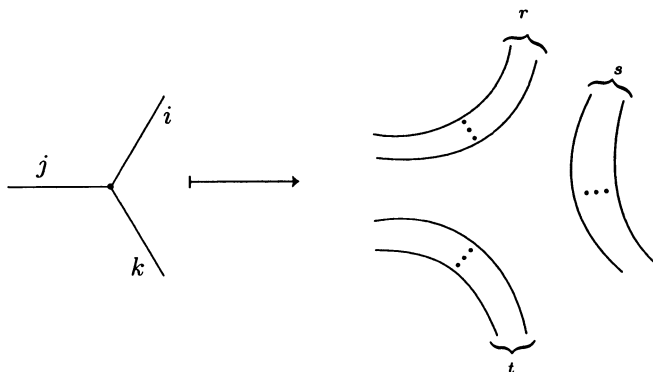


Figure 4.3

The remarks made in Section 4.1 directly extend to graph-nets.

The next theorem describes the behavior of the bracket function under the basic moves. We shall use the following notation: for  $x \in \mathbb{C}$  and non-negative integers  $m, n$  set

$$\pi_{m,n}(x) = \prod_{a=1}^m \prod_{b=1}^n \left[ x + \frac{n-m}{2} + a - b \right] \in \mathbb{C}.$$

Note that

$$\pi_{m,n}(x) = \pi_{n,m}(x) = (-1)^{mn} \pi_{m,n}(-x) = (-1)^{mn} \pi_{n,m}(-x).$$

**4.3. Theorem.** *Let  $L_i$  be a graph-net obtained from a graph-net  $L$  by the transformation  $\omega_i$  with  $i = 0, 1, 2, 3$ . Let  $m, n$  be the colors of the two arcs of  $L$  undergoing the moves  $\omega_0, \omega_2$ . Let  $m$  be the color of the arc of  $L$  undergoing  $\omega_1$ . We have*

$$\langle L_0 \rangle = (-1)^{mn} \langle L \rangle, \quad (4.3.a)$$

$$\langle L_1 \rangle = (-1)^m \frac{[x - m - 1]}{[x - 1]} \pi_{m,m}(x - 1) \langle L \rangle, \quad (4.3.b)$$

$$\langle L_2 \rangle = (-1)^{mn} \pi_{m,n}(x + 1) \pi_{m,n}(x - 1) \langle L \rangle, \quad (4.3.c)$$

$$\langle L_3 \rangle = \langle L \rangle. \quad (4.3.d)$$

Theorem 4.3 is proven in Sections 5 and 7.5.

#### 4.4. Mirror images

For any graph-net  $L$  with 0 inputs and 0 outputs we can consider its mirror image  $\text{mir}(L)$  with respect to any line in  $\mathbb{R}^2$ . The distinguished region attached

to a crossing in  $\text{mir}(L)$  is the mirror image of the distinguished region attached to the corresponding crossing of  $L$ . The formal angles are preserved under the passage to the mirror image. Any coloring of  $L$  induces a coloring of  $\text{mir}(L)$  in the obvious way. It turns out that for any colored graph-net  $L$  with 0 inputs and 0 outputs, we have

$$\langle L \rangle = \langle \text{mir}(L) \rangle. \quad (4.4.a)$$

For uncolored nets, this equality follows from the invariance of Figure 2.1 under the mirror symmetry with respect to the vertical line. For colored nets, (4.4.a) follows from the fact that the cabling of nets commutes with the passage to the mirror image:  $(\text{mir}(L))^c = \text{mir}(L^c)$ .

#### 4.5. Comparison with Kauffman's bracket polynomial

Observe first that the bracket function of a net (or a graph-net) does not change if we replace  $q$  with  $q^{-1}$ . In this subsection we assume that  $|q| < 1$ .

The main ingredient of the bracket function for nets is the  $R$ -matrix  $R(x) = [x]e_1 + [x-1]1_2 \in TL_2$ . Here  $1_2, e_1$  are the generators of the Temperley-Lieb algebra  $TL_2$  defined in Section 2.4 and  $R(x) \in TL_2$ . Restricting ourselves to  $x \in \mathbb{R}$  we easily compute that

$$q^{-1/2}(q^{-1} - q) \lim_{x \rightarrow +\infty} q^x R(x) = q^{-1/2}e_1 + q^{1/2}1_2. \quad (4.5.a)$$

The right-hand side is the  $R$ -matrix used by Kauffman in his definition of the bracket polynomial of a framed link. We conclude that when all formal angles of a net  $L$  converge to  $+\infty$ , the bracket function  $\langle L \rangle$  (multiplied by appropriate powers of  $q$  and  $q^{-1} - q$ ) converges to the value in  $q^{1/2}$  of the bracket polynomial of the link obtained by replacing all crossings of  $L$  in the way indicated in Figure 4.4.

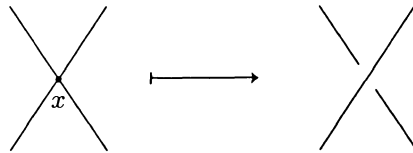


Figure 4.4

These observations easily extend to colored nets and graph-nets. To see this, denote by  $L_{m,n}(x)$  the relative  $(m+n, m+n)$ -net shown on the right of Figure 4.2 with  $x_{a,b}$  given by (4.1.a). Set

$$R_{m,n}(x) = \langle L_{m,n}(x) \rangle \in TL_{m+n}.$$



Applying formula (4.5.a) to all crossings of  $L_{m,n}(x)$  we obtain

$$q^{-mn/2}(q^{-1} - q)^{mn} \lim_{x \rightarrow +\infty} q^{mnx} R_{m,n}(x) = R_{m,n} \quad (4.5.b)$$

where  $R_{m,n} \in TL_{m+n}$  corresponds in Kauffman's model to the tangle obtained from this net by replacing all crossings as in Figure 4.4. Thus, when all formal angles of a colored graph-net converge to  $+\infty$ , the bracket function (multiplied by appropriate powers of  $q$  and  $q^{-1} - q$ ) converges to the value in  $q^{1/2}$  of the bracket polynomial of the colored graph obtained by replacing all crossings as in Figure 4.4.

Similar computations hold for  $x \rightarrow -\infty$ :

$$q^{1/2}(q - q^{-1}) \lim_{x \rightarrow -\infty} q^{-x} R(x) = q^{1/2}e_1 + q^{-1/2}1_2 \quad (4.5.c)$$

$$q^{mn/2}(q - q^{-1})^{mn} \lim_{x \rightarrow -\infty} q^{-mnx} R_{m,n}(x) = R_{n,m}^{-1}. \quad (4.5.d)$$

## 5. Proof of Theorem 4.3

In this section we prove equalities (4.3.a), (4.3.c), (4.3.d). The proof of (4.3.b) is postponed to Section 7.5.

### 5.1. Notation

Consider the  $(m+n, m+n)$ -net drawn on the right in Figure 4.2. We shall consider 4 different choices of the numbers  $\{x_{a,b}\}$ :

$$\begin{aligned} x_{a,b} &= x + \frac{n-m}{2} + a - b, \\ x_{a,b} &= x + \frac{m+n}{2} + 1 - a - b, \\ x_{a,b} &= x + \frac{m-n}{2} + b - a, \\ x_{a,b} &= x - \frac{m+n}{2} - 1 + a + b. \end{aligned}$$

Denote the resulting (uncolored) relative nets by  $X_{m,n}(x)$ ,  $Y_{m,n}(x)$ ,  $Z_{m,n}(x)$ , and  $T_{m,n}(x)$ , respectively.

We shall write  $L_1 \doteq L_2$  for colored nets (or graph-nets)  $L_1, L_2$  such that  $\langle L_1 \rangle = \langle L_2 \rangle$ . Similarly, for a colored graph-net  $L$  and a morphism  $f$  in the Temperley-Lieb category, we write  $L \doteq f$  and  $f \doteq L$  whenever  $f = \langle L \rangle$ .

As above, by the symbol  $1_m$  we denote the  $(m, m)$ -net consisting of  $m$  embedded disjoint vertical intervals. Recall the  $(m, m)$ -net  $\tilde{F}_m$  defined in Section 2.8 (up to moves  $\omega_3, \omega_3^{-1}$ ).

**5.2. Lemma.** *We have the following equivalences modulo  $\omega_3, \omega_3^{-1}$ :*

$$(\tilde{F}_n \otimes 1_m)X_{m,n}(x) \sim T_{m,n}(x)(1_m \otimes \tilde{F}_n), \quad (5.2.a)$$

$$(\tilde{F}_n \otimes 1_m)Y_{m,n}(x) \sim Z_{m,n}(x)(1_m \otimes \tilde{F}_n), \quad (5.2.b)$$

$$(1_n \otimes \tilde{F}_m)X_{m,n}(x) \sim Y_{m,n}(x)(\tilde{F}_m \otimes 1_n). \quad (5.2.c)$$

$$(1_n \otimes \tilde{F}_m)T_{m,n}(x) \sim Z_{m,n}(x)(\tilde{F}_m \otimes 1_n). \quad (5.2.d)$$

Here the symbol  $\sim$  denotes equivalences modulo the moves  $\omega_3, \omega_3^{-1}$ . Note that the net  $(\tilde{F}_n \otimes 1_m)X_{m,n}(x)$  is obtained by sewing  $\tilde{F}_n$  to the left upper bunch of  $n$  strings of  $X_{m,n}(x)$ , the net  $T_{m,n}(x)(1_m \otimes \tilde{F}_n)$  is obtained by sewing  $\tilde{F}_n$  to the right lower bunch of  $n$  strings of  $T_{m,n}(x)$ , etc.

**Proof of Lemma.** For  $m = 1$ , equivalence (5.2.a) is straightforward, one has to apply the move  $\omega_3$  exactly  $n(n-1)/2$  times consecutively pushing down the  $n(n-1)/2$  crossing points of  $\tilde{F}_n$ . The general case follows by induction on  $m$ . The proof of (5.2.b) – (5.2.d) is similar.

**5.3. Lemma.**

$$(F_n \otimes 1_m)X_{m,n}(x) \doteq T_{m,n}(x)(1_m \otimes F_n), \quad (5.3.a)$$

$$(F_n \otimes 1_m)Y_{m,n}(x) \doteq Z_{m,n}(x)(1_m \otimes F_n), \quad (5.3.b)$$

$$(1_n \otimes F_m)X_{m,n}(x) \doteq Y_{m,n}(x)(F_m \otimes 1_n). \quad (5.3.c)$$

$$(1_n \otimes F_m)T_{m,n}(x) \doteq Z_{m,n}(x)(F_m \otimes 1_n). \quad (5.3.d)$$

**Proof.** Equality (5.3a) is obtained from (5.2a) by taking the bracket function and dividing by  $\prod_{1 \leq i < j \leq n} [i - j - 1]$ . Equalities (5.3b)–(5.3d) are obtained similarly from (5.2.b)–(5.2.d).

**5.4. Proof of (4.3a).** It is enough to consider the case where  $L, L_0$  are the  $(2,2)$ -nets drawn in Figure 0.2 on the left and on the right of the arrow. Let  $m, n$  be the colors of the branches of  $L$  in the order of their bottom ends countered from left to right. By definition,

$$L^c = (F_n \otimes F_m)X_{m,n}(x)(F_m \otimes F_n).$$

Similarly,

$$L_2^c = (F_n \otimes F_m)Z(F_m \otimes F_n)$$

where  $Z$  is the net obtained from  $X_{n,m}(1-x)$  by rotation in the counter-clockwise direction to the angle of  $90^\circ$ . Applying to all  $mn$  crossings of  $Z$  the transformation  $\omega_0^{-1}$  we obtain the net  $Y_{m,n}(x)$ . By (2.6.a), (3.1.a), and Lemma 5.3,

$$\begin{aligned} \langle L_2^c \rangle &= (f_n \otimes f_m) \langle Z \rangle (f_m \otimes f_n) = (-1)^{mn} (f_n \otimes f_m) \langle Y_{m,n}(x) \rangle (f_m \otimes f_n) \\ &= (-1)^{mn} (f_n \otimes f_m) \langle Y_{m,n}(x) \rangle (f_m \otimes 1_n) (f_m \otimes f_n) \\ &= (-1)^{mn} (f_n \otimes f_m) (1_n \otimes f_m) \langle X_{m,n}(x) \rangle (f_m \otimes f_n) \\ &= (-1)^{mn} (f_n \otimes f_m) \langle X_{m,n}(x) \rangle (f_m \otimes f_n) = (-1)^{mn} \langle L^c \rangle. \end{aligned}$$

### 5.5. Lemma.

$$(F_n \otimes F_m)X_{m,n}(x)(F_m \otimes F_n) \doteq (F_n \otimes F_m)X_{m,n}(x).$$

**Proof.** By (5.3a) and (5.3d),

$$(F_n \otimes F_m)X_{m,n}(x) \doteq Z_{m,n}(x)(F_m \otimes F_n).$$

Similarly,

$$(F_n \otimes F_m)X_{m,n}(x)(F_m \otimes F_n) \doteq Z_{m,n}(x)(F_m^2 \otimes F_n^2) \doteq Z_{m,n}(x)(F_m \otimes F_n).$$

**5.6. Proof of (4.3c).** It suffices to consider the case where  $L$  consists of two vertical intervals colored with  $m, n$  and where  $L_2$  is obtained from  $L$  by  $\omega_2$  as in Figure 1.1. By definition,  $L^c = F_m \otimes F_n$  and

$$L_2^c = (F_m \otimes F_n)X_{n,m}(-x)(F_n \otimes F_m)X_{m,n}(x)(F_m \otimes F_n).$$

It follows from Lemma 5.5 that the term  $F_n \otimes F_m$  may be removed from this expression without changing the bracket. Hence

$$\langle L_2^c \rangle = (f_m \otimes f_n) \langle X_{n,m}(-x)X_{m,n}(x) \rangle (f_m \otimes f_n).$$

It remains to observe that the net  $X_{n,m}(-x)X_{m,n}(x)$  may be transformed into  $1_m \otimes 1_n$  applying  $mn$  times the move  $\omega_2^{-1}$ . Each such application cancels the

$(a, b)$ -th crossing of  $X_{m,n}(x)$  with the  $(b, a)$ -th crossing of  $X_{n,m}(-x)$ . The sum of the corresponding formal angles equals

$$(x + \frac{n-m}{2} + a - b) + (-x + \frac{m-n}{2} + b - a) = 0$$

which allows us to apply (2.6.c). We obtain

$$\begin{aligned} \langle L_2^c \rangle &= (-1)^{mn} \pi_{m,n}(x+1) \pi_{m,n}(x-1) (f_m \otimes f_n) \\ &= (-1)^{mn} \pi_{m,n}(x+1) \pi_{m,n}(x-1) \langle L^c \rangle. \end{aligned}$$

**5.7. Proof of (4.3d).** It is enough to consider the case where  $L, L_3$  are the  $(3,3)$ -nets drawn in Figure 1.1 on the left and on the right of the arrow  $\omega_3$ . Let  $m, n, r$  be the colors of the three branches of  $L$  in the order of their bottom ends countered from left to right. By definition,

$$\begin{aligned} L^c &= (F_r \otimes F_n \otimes F_m)(X_{n,r}(y) \otimes 1_m)(F_n \otimes F_r \otimes F_m) \\ &\quad \circ (1_n \otimes X_{m,r}(z))(F_n \otimes F_m \otimes F_r)(X_{m,n}(x) \otimes 1_r)(F_m \otimes F_n \otimes F_r). \end{aligned}$$

It follows from Lemma 5.5 that the terms  $F_n \otimes F_r \otimes F_m$  and  $F_n \otimes F_m \otimes F_r$  may be removed from this expression without changing the bracket. Hence

$$L^c \doteq (F_r \otimes F_n \otimes F_m)(X_{n,r}(y) \otimes 1_m)(1_n \otimes X_{m,r}(z))(X_{m,n}(x) \otimes 1_r)(F_m \otimes F_n \otimes F_r).$$

Similar arguments show that

$$L_3^c \doteq (F_r \otimes F_n \otimes F_m)(1_r \otimes X_{m,n}(x))(X_{m,r}(z) \otimes 1_n)(1_m \otimes X_{n,r}(y))(F_m \otimes F_n \otimes F_r).$$

Therefore equality (4.3.d) follows from (2.6.d) and the obvious fact that for  $z = x + y$ , the net  $(X_{n,r}(y) \otimes 1_m)(1_n \otimes X_{m,r}(z))(X_{m,n}(x) \otimes 1_r)$  may be transformed into  $(1_r \otimes X_{m,n}(x))(X_{m,r}(z) \otimes 1_n)(1_m \otimes X_{n,r}(y))$  by  $\omega_3$ -moves.

**5.8. Remark.** In the definition of cabling for colored nets we have replaced all crossings with nets of type  $X_{m,n}(x)$ . Instead we can use nets of types  $Y_{m,n}(x), Z_{m,n}(x), T_{m,n}(x)$  without changing the bracket function. This follows from the equalities

$$\begin{aligned} (F_n \otimes F_m)X_{m,n}(x)(F_m \otimes F_n) &\doteq (F_n \otimes F_m)Y_{m,n}(x)(F_m \otimes F_n) \\ &\doteq (F_n \otimes F_m)Z_{m,n}(x)(F_m \otimes F_n) \\ &\doteq (F_n \otimes F_m)T_{m,n}(x)(F_m \otimes F_n). \end{aligned}$$

These equalities follow from Lemma 5.3 and the identity  $\langle F_k^2 \rangle = \langle F_k \rangle = f_k$ .

## 6. Trigonometric $6j$ -symbols

Throughout this section  $q \neq 0$  is not a root of unity.

### 6.1. Preliminaries on $\theta$ -nets

The  $\theta$ -net is the embedded colored graph-net in  $\mathbb{R}^2$  with 2 vertices connected by three embedded edges colored with  $i, j, k$ ; see Figure 6.1. This colored net is defined only when the triple  $(i, j, k)$  is admissible in the sense of Section 4.2. The bracket function of this net is denoted by  $\theta(i, j, k)$ . It is easy to observe that  $\theta(i, j, k)$  is an invariant of *unordered* triple  $i, j, k$ . The following explicit formula for  $\theta(i, j, k)$  is verified, for instance, in [KL, Chapter 6]:

$$\theta(i, j, k) = (-1)^{(i+j+k)/2} \frac{[\frac{i+j+k}{2} + 1]! [\frac{i+j-k}{2}]! [\frac{i+k-j}{2}]! [\frac{j+k-i}{2}]!}{[i]! [j]! [k]!}. \quad (6.1.a)$$

Here for a positive integer  $a$  we have  $[a]! = [1][2]\dots[a]$  and for  $a = 0$  we have  $[a]! = 1$ . Since  $q$  is not a root of unity,  $\theta(i, j, k) \neq 0$ .

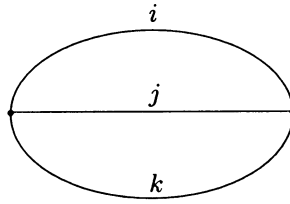


Figure 6.1

In the case  $k = 0$  the  $\theta$ -net in Figure 6.1 is just an embedded circle colored with  $i = j$ . Its bracket function is denoted by  $\Delta(i)$ . We have

$$\Delta(i) = \theta(i, i, 0) = (-1)^i [i + 1].$$

We fix for any unordered admissible triple  $(i, j, k)$  a square root  $\tilde{\theta}(i, j, k) \in \mathbb{C}$  of  $\theta(i, j, k)$ .

### 6.2. Preliminaries on standard $6j$ -symbols

Let  $i, j, k, l, m, n$  be nonnegative integers such that the triples  $(i, j, n), (i, l, m), (j, k, m), (k, l, n)$  are admissible. The corresponding  $6j$ -symbol (without spectral parameter) is a number

$$\left\{ \begin{array}{ccc} i & j & n \\ k & l & m \end{array} \right\} \in \mathbb{C} \quad (6.2.a)$$

defined by the formula in Figure 6.2. (The second equality in Figure 6.2 follows from the fact that the two graph-nets are ambiently isotopic in the 2-sphere  $S^2$ .) The  $6j$ -symbol (6.2.a) is computed in a different way in Figure 6.3 where  $q_a = \sqrt{-1}^a q^{a(a+2)/4}$  for an integer  $a$ . Here the crossing in the first picture stands for  $R_{m,n} \in TL_{m+n}$  as defined in Section 4.5, the crossing in the second picture stands for  $R_{n,m}^{-1} \in TL_{m+n}$ .

$$\left\{ \begin{matrix} i & j & n \\ k & l & m \end{matrix} \right\} = \left\langle \begin{array}{c} \text{graph-net with crossing } R_{m,n} \end{array} \right\rangle = \left\langle \begin{array}{c} \text{graph-net with crossing } R_{n,m}^{-1} \end{array} \right\rangle$$

Figure 6.2

$$\begin{aligned} \left\{ \begin{matrix} i & j & n \\ k & l & m \end{matrix} \right\} &= q_i q_j^{-1} q_k q_l^{-1} \left\langle \begin{array}{c} \text{graph-net with crossing } R_{m,n} \end{array} \right\rangle = \\ &= q_i^{-1} q_j q_k^{-1} q_l \left\langle \begin{array}{c} \text{graph-net with crossing } R_{n,m}^{-1} \end{array} \right\rangle \end{aligned}$$

Figure 6.3

The  $6j$ -symbol (6.2.a) has a full tetrahedral symmetry, i.e., it is invariant under arbitrary permutations of columns in the matrix

$$\begin{bmatrix} i & j & n \\ k & l & m \end{bmatrix} \quad (6.2.b)$$

and under the permutation of rows in any  $(2 \times 2)$ -submatrix.

There is an explicit formula expressing the  $6j$ -symbol (6.2.a) via its 6 in-

teger parameters (see [MV]). In particular, in the case  $n = k + l$  we have

$$\begin{aligned} \left\{ \begin{matrix} i & j & n \\ k & l & m \end{matrix} \right\} &= \left\{ \begin{matrix} k & l & n \\ i & j & m \end{matrix} \right\} \\ &= (-1)^{(i+j+n)/2} \left[ \frac{i+j+n}{2} + 1 \right]! \frac{\left[ \frac{i-j+n}{2} \right]! \left[ \frac{-i+j+n}{2} \right]! \left[ \frac{m+i-l}{2} \right]! \left[ \frac{m+j-k}{2} \right]!}{[i]! [j]! [m]! [n]!}. \end{aligned} \quad (6.2.c)$$

One can consider various versions of the  $6j$ -symbols (without spectral parameter) which differ in a normalization factor. Besides the  $6j$ -symbol (6.2.a), we shall consider its renormalization defined by

$$\left| \begin{matrix} i & j & n \\ k & l & m \end{matrix} \right| = \frac{\left\{ \begin{matrix} i & j & n \\ k & l & m \end{matrix} \right\}}{\bar{\theta}(i, j, n) \bar{\theta}(i, l, m) \bar{\theta}(j, k, m) \bar{\theta}(k, l, n)} \in \mathbb{C}.$$

This  $6j$ -symbol has the same tetrahedral symmetry as (6.2.a). The normalization factor  $(\bar{\theta}(i, j, n) \bar{\theta}(i, l, m) \bar{\theta}(j, k, m) \bar{\theta}(k, l, n))^{-1}$  allows to simplify the properties of the  $6j$ -symbols; for details see [KR], [KL], [Tu].

### 6.3. Definition of $6j$ -symbols with spectral parameter

Let  $i, j, k, l, m, n$  be nonnegative integers such that the triples  $(i, j, n), (i, l, m), (j, k, m), (k, l, n)$  are admissible. For any  $x \in \mathbb{C}$ , we define the trigonometric  $6j$ -symbol with spectral parameter  $x$

$$\left\{ \begin{matrix} i & j & n \\ k & l & m \end{matrix} \right\} (x) \in \mathbb{C} \quad (6.3.a)$$

by the formula in Figure 6.4. This  $6j$ -symbol is the bracket of the colored graph-net in Figure 6.4 which has 4 trivalent vertices and one crossing with formal angle  $x$ .

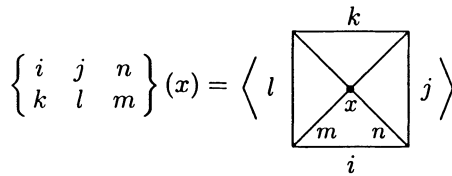


Figure 6.4

Note that

$$\left\{ \begin{matrix} i & j & 0 \\ k & l & m \end{matrix} \right\} (x) = \left\{ \begin{matrix} i & j & 0 \\ k & l & m \end{matrix} \right\} = \theta(i, k, m) \delta_i^k \delta_i^j \quad (6.3.b)$$

and

$$\left\{ \begin{matrix} i & j & n \\ k & l & 0 \end{matrix} \right\} (x) = \left\{ \begin{matrix} i & j & n \\ k & l & 0 \end{matrix} \right\} = \theta(i, j, n) \delta_i^k \delta_j^l. \quad (6.3.c)$$

A convenient renormalization of the trigonometric  $6j$ -symbol with spectral parameter  $x$  is defined by

$$\left| \begin{matrix} i & j & n \\ k & l & m \end{matrix} \right| (x) = \frac{\left\{ \begin{matrix} i & j & n \\ k & l & m \end{matrix} \right\} (x)}{\tilde{\theta}(i, j, n) \tilde{\theta}(i, l, m) \tilde{\theta}(j, k, m) \tilde{\theta}(k, l, n)}. \quad (6.3.d)$$

It follows from formula (4.5.b) and the first equality in Figure 6.3 that for  $|q| < 1$ ,

$$\lim_{x \rightarrow +\infty} q^{mnx} \left| \begin{matrix} i & j & n \\ k & l & m \end{matrix} \right| (x) = q^{mn/2} (q^{-1} - q)^{-mn} q_i^{-1} q_j q_k^{-1} q_l \left| \begin{matrix} i & j & n \\ k & l & m \end{matrix} \right|.$$

Similarly,

$$\lim_{x \rightarrow -\infty} q^{-mnx} \left| \begin{matrix} i & j & n \\ k & l & m \end{matrix} \right| (x) = q^{-mn/2} (q - q^{-1})^{-mn} q_i q_j^{-1} q_k q_l^{-1} \left| \begin{matrix} i & j & n \\ k & l & m \end{matrix} \right|.$$

As an exercise, the reader may check that these formulas are compatible with the symmetries established in the next theorem.

**6.3.1. Theorem.** *The trigonometric  $6j$ -symbols (6.3.a) and (6.3.d) are invariant under the permutation of rows in any  $(2 \times 2)$ -submatrix of the matrix (6.2.b).*

**Proof.** It suffices to consider one of these two  $6j$ -symbols because the product  $\tilde{\theta}(i, j, n) \tilde{\theta}(i, l, m) \tilde{\theta}(j, k, m) \tilde{\theta}(k, l, n)$  is invariant under the permutations in question. It follows from (4.3.a) that

$$\left| \begin{matrix} i & j & n \\ k & l & m \end{matrix} \right| (x) = (-1)^{mn} \left| \begin{matrix} j & k & m \\ l & i & n \end{matrix} \right| (1-x). \quad (6.3.e)$$

Applying this formula twice we obtain

$$\left| \begin{matrix} i & j & n \\ k & l & m \end{matrix} \right| (x) = \left| \begin{matrix} k & l & n \\ i & j & m \end{matrix} \right| (x). \quad (6.3.f)$$

Taking the mirror image of the graph-net in Figure 6.4 with respect to the vertical axis we obtain the same graph-net with  $i, j, k, l, m, n$  replaced by



$i, l, k, j, n, m$ , respectively. It follows from (4.5.a) that

$$\begin{vmatrix} i & j & n \\ k & l & m \end{vmatrix} (x) = \begin{vmatrix} i & l & m \\ k & j & n \end{vmatrix} (x). \quad (6.3.g)$$

**6.4. Theorem (the orthonormality relation).** *For any nonnegative integers  $i, i', j, l, m, n$ , such that the triples  $(i, j, n), (i, l, m), (i', j, n), (i', l, m)$  are admissible and any  $x \in \mathbb{C}$ , we have*

$$\begin{aligned} \Delta(i) \sum_r \Delta(r) \begin{vmatrix} i & j & n \\ r & l & m \end{vmatrix} (x) \begin{vmatrix} i' & j & n \\ r & l & m \end{vmatrix} (-x) \\ = (-1)^{mn} \pi_{m,n}(x+1) \pi_{m,n}(x-1) \delta_{i'}^i. \end{aligned}$$

It is understood that we sum up over all nonnegative integers  $r$  such that the triples  $(j, m, r), (l, n, r)$  are admissible.

The proof of Theorem 6.4 is given in Section 6.8.

**6.5. Theorem (the Yang-Baxter identity).** *For any nonnegative integers  $j_1, j_2, \dots, j_9$  and any  $x, y \in \mathbb{C}$ ,*

$$\begin{aligned} \sum_r \Delta(r) \begin{vmatrix} j_1 & j_2 & j_8 \\ r & j_6 & j_7 \end{vmatrix} (x) \begin{vmatrix} j_2 & j_3 & j_9 \\ j_4 & r & j_7 \end{vmatrix} (x+y) \begin{vmatrix} r & j_4 & j_9 \\ j_5 & j_6 & j_8 \end{vmatrix} (y) \\ = \sum_r \Delta(r) \begin{vmatrix} j_2 & j_3 & j_9 \\ r & j_1 & j_8 \end{vmatrix} (y) \begin{vmatrix} j_1 & r & j_9 \\ j_5 & j_6 & j_7 \end{vmatrix} (x+y) \begin{vmatrix} r & j_3 & j_8 \\ j_4 & j_5 & j_7 \end{vmatrix} (x). \end{aligned} \quad (6.5.a)$$

It is understood that we sum up over all nonnegative integers  $r$  such that the  $6j$ -symbols involved in the formula are well defined.

Theorems 6.3.1, 6.4, and 6.5 are analogous to the properties of the elliptic  $6j$ -symbols established in [DJMO]. (Exact relationships between trigonometric and elliptic  $6j$ -symbols will be discussed in [FT].)

Our proof of Theorems 6.4 and 6.5 is based on a geometric approach. The proof is given in Sections 6.8 and 6.9 using the following two lemmas.

**6.6. Lemma.** *For any colored graph-net  $L$  with one  $i$ -colored input and one  $j$ -colored output, we have*

$$\langle L \rangle = \frac{\langle L' \rangle}{\Delta(i)} \delta_j^i f_i \in TL_i \quad (6.6.a)$$

where  $L'$  is the colored graph-net obtained by closing the ends of  $L$ .

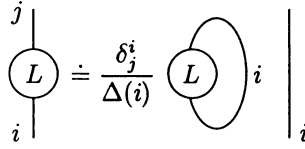


Figure 6.5

A graphical version of formula (6.6.a) is given in Figure 6.5 where  $L$  is schematically represented by a disk with the symbol  $L$  inside.

**Proof of Lemma.** Note that  $\langle L \rangle = f_j t f_i$  where  $t \in TL_{i,j}$  is a linear combination of embedded systems of arcs with  $i$  inputs and  $j$  outputs. We claim that if  $i > j$  then for every such system of arcs,  $t'$ , we have  $t' f_i = 0$ . Indeed,  $t'$  contains an arc joining two inputs. A minimal arc of such kind connects two consecutive inputs. Hence  $t' = t'' e_k$  where  $e_k$  is one of the generators of  $TL_i$  with  $k < i$ . Therefore  $t' f_i = t'' e_k f_i = 0$ . This implies that  $f_j t f_i = 0$  for  $i > j$ . Similarly,  $f_j t f_i = 0$  for  $i < j$ .

Let  $i = j$ . Then  $\langle L \rangle = f_i t f_i = C f_i$  with  $C \in \mathbb{C}$ . To compute  $C$ , observe that  $\langle L' \rangle = C \langle S_i \rangle = C \Delta(i)$  where  $S_i$  is an embedded  $i$ -colored circle in  $\mathbb{R}^2$ . Therefore  $C = (\Delta(i))^{-1} \langle L' \rangle$ .

**6.7. Lemma.** For any nonnegative integers  $m, n$ , we have the equality in Figure 6.6.

Figure 6.6

**Proof of Lemma.** It follows from the properties of the Jones-Wenzl idempotents that for any integers  $m, n, r \geq 0$  the vector space  $(f_m \otimes f_n) TL_{r, m+n} f_r$  is one-dimensional if the triple  $(m, n, r)$  is admissible and zero otherwise. For an admissible triple  $(m, n, r)$ , this vector space has a natural generator  $a_r^{m,n}$  defined in Figure 6.7. (Thus,  $a_r^{m,n} \in TL_{r, m+n}$  is the bracket function of the simplest graph-net with one vertex, one  $r$ -colored input and two outputs colored with  $m, n$ .) Similarly, the vector space  $f_r TL_{m+n, r} (f_m \otimes f_n)$  is one-dimensional if the triple  $(m, n, r)$  is admissible and zero otherwise. For an admissible triple  $(m, n, r)$ , this vector space has a natural generator  $a_{m,n}^r$  defined in Figure 6.7.

It is known (see e.g. [Tu, Chapter XII]) that the idempotents  $f_1 \in TL_1, f_2 \in TL_2, \dots$  generate the set of morphisms of the Temperley-Lieb cat-

$$a_r^{m,n} \doteq \begin{array}{c} m \quad n \\ \diagdown \quad \diagup \\ \quad \quad r \end{array} \quad a_{m,n}^r \doteq \begin{array}{c} r \\ \diagup \quad \diagdown \\ m \quad n \end{array}$$

Figure 6.7

egory as a two-sided ideal over  $\mathbb{C}$ . This means that every vector space  $TL_{i,j}$  is generated by vectors of type  $xf_r y$  with  $r \geq 0, y \in TL_{i,r}, x \in TL_{r,j}$ . In particular,  $1_m \otimes 1_n \in TL_{m+n}$  may be presented as a linear combination (with complex coefficients) of such vectors. Multiplying such a decomposition of  $1_m \otimes 1_n$  from the left and the right by  $f_m \otimes f_n$  and using the results of the previous paragraph we obtain

$$f_m \otimes f_n = \sum_r c_{m,n}^r a_r^{m,n} a_{m,n}^r \quad (6.7.a)$$

where  $c_{m,n}^r \in \mathbb{C}$  and  $r$  runs over nonnegative integers such that the triple  $(m, n, r)$  is admissible. To complete the proof of the lemma it suffices to show that  $c_{m,n}^r = \Delta(r)(\theta(m, n, r))^{-1}$ .

For any  $r, r_0$  such that the triples  $(m, n, r), (m, n, r_0)$  are admissible, Lemma 6.6 implies that

$$a_{m,n}^{r_0} a_r^{m,n} = \delta_r^{r_0} (\Delta(r))^{-1} \theta(m, n, r_0).$$

It is clear that  $a_{m,n}^{r_0} (f_m \otimes f_n) = a_{m,n}^{r_0}$ . Therefore multiplying (6.7.a) by  $a_{m,n}^{r_0}$  from the left we obtain

$$a_{m,n}^{r_0} = c_{m,n}^{r_0} (\Delta(r_0))^{-1} \theta(m, n, r_0) a_{m,n}^{r_0}.$$

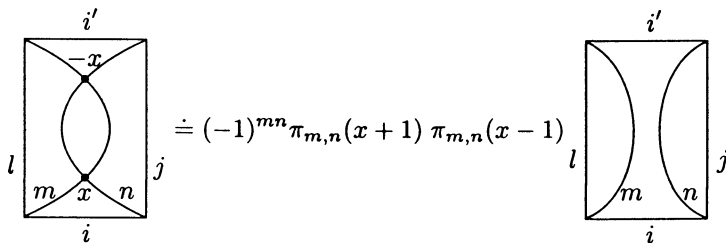
Hence  $c_{m,n}^{r_0} = \Delta(r_0)(\theta(m, n, r_0))^{-1}$ .

**6.8. Proof of Theorem 6.4.** We compute in two different ways the bracket function of the graph-net,  $L$ , shown on the left in Figure 6.8. The first computation given in Figure 6.8 uses (4.3.c). The bracket on the right-hand side can be computed using Lemma 6.6. We obtain

$$\langle L \rangle = (-1)^{mn} \pi_{m,n}(x+1) \pi_{m,n}(x-1) \theta(i, j, n) \theta(i, l, m) (\Delta(i))^{-1} \delta_i^i.$$

A second computation of  $\langle L \rangle$  is given in Figure 6.9. Here we apply Lemma 6.7 to branches connected by the dotted lines and then apply Lemma 6.6. We obtain

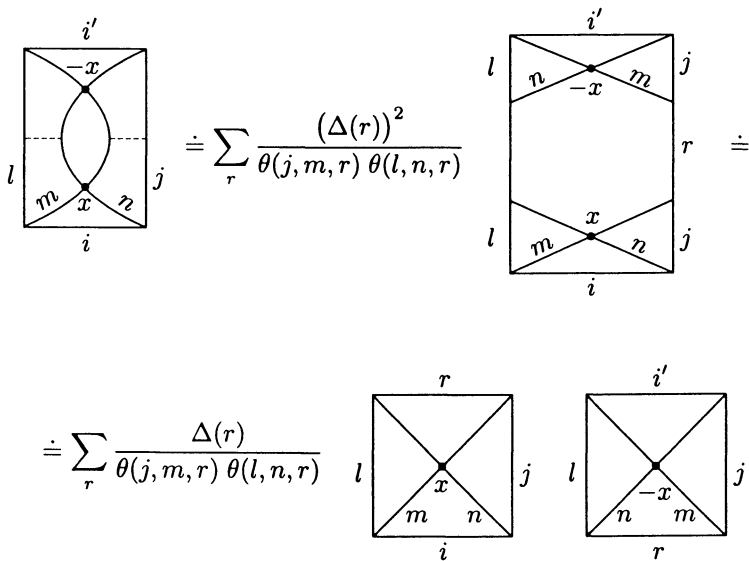
$$\langle L \rangle = \sum_{r \geq 0} \Delta(r) (\theta(j, m, r) \theta(l, n, r))^{-1} \left\{ \begin{array}{ccc} i & j & n \\ r & l & m \end{array} \right\} (x) \left\{ \begin{array}{ccc} r & j & m \\ i' & l & n \end{array} \right\} (-x).$$

**Figure 6.8**

Comparing two expressions for  $\langle L \rangle$  and using the symmetry

$$\left| \begin{array}{ccc} r & j & m \\ i' & l & n \end{array} \right| (-x) = \left| \begin{array}{ccc} i' & j & n \\ r & l & m \end{array} \right| (-x)$$

(which follows from the results of Section 6.3) we obtain the orthonormality relation stated in Theorem 6.4.

**Figure 6.9**

**6.9. Proof of Theorem 6.5.** Proof of Theorem 6.5 uses the same idea as the proof of Theorem 6.4.

Set

$$\Theta = \tilde{\theta}(j_1, j_2, j_8) \tilde{\theta}(j_2, j_3, j_9) \tilde{\theta}(j_3, j_4, j_7) \tilde{\theta}(j_4, j_5, j_8) \tilde{\theta}(j_5, j_6, j_9) \tilde{\theta}(j_6, j_1, j_7) \in \mathbb{C}.$$

Figure 1 shows two circular diagrams,  $L$  and  $L'$ , representing a 2D lattice. Diagram  $L$  (left) has vertices  $x, y, z$  and edges  $j_1, j_2, j_3, j_4, j_5, j_6, j_7, j_8, j_9$ . Diagram  $L'$  (right) has vertices  $x, y, z$  and edges  $j_1, j_2, j_3, j_4, j_5, j_6, j_7, j_8, j_9$ . The diagrams are related by a duality transformation, indicated by a triple bar symbol between them.

**Figure 6.10**

It was shown by Kirillov and Reshetikhin [KR] that the Jones polynomial of links may be computed in terms of face models and  $6j$ -symbols. A similar model yields the bracket function of a colored graph-net  $L \subset \mathbb{R}^2$  as follows. Consider an assignment of a nonnegative integer, the color, to every component of  $\mathbb{R}^2 \setminus L$  such that (i) the unbounded region is colored with 0; (ii) if  $i$  is a color of an edge of  $L$  and  $j, k$  are the colors of two adjacent regions, then the triple  $(i, j, k)$  is admissible. It is obvious that the set of such colorings is finite, it is denoted by  $\text{col}(L)$ . Let  $\varphi \in \text{col}(L)$ . For a crossing  $\alpha$  of  $L$ , set

$$|\alpha|_\varphi = \left| \begin{array}{ccc} i & j & n \\ k & l & m \end{array} \right| (x)$$

$$|\beta|_\varphi = \begin{vmatrix} i & j & n \\ k & l & m \end{vmatrix}$$

where  $i, j, k, l, m, n$  are the colors of edges of  $L$  and components of  $\mathbb{R}^2 \setminus L$  as in

Figure 6.11. As in the classical theory, we have

$$\langle L \rangle = \sum_{\varphi \in \text{col}(L)} \prod_{\alpha} |\alpha|_{\varphi} \prod_{\beta} |\beta|_{\varphi} \prod_X \Delta(\varphi(X))^{\chi(X)}$$

where  $\alpha$  runs over crossings of  $L$ ,  $\beta$  runs over trivalent vertices of  $L$ ,  $X$  runs over components of  $\mathbb{R}^2 \setminus L$ , and  $\chi$  is the Euler characteristic. The proof of this formula is essentially the same as in the classical case (see e.g. [KL, Chapter 11]).

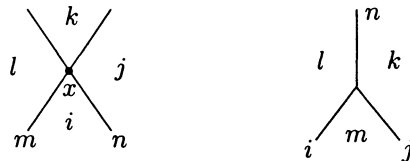


Figure 6.11

## 7. Tetrahedral symmetry of the trigonometric $6j$ -symbols

Throughout this section we assume that  $q$  is not a root of unity.

### 7.1. Notation

For  $x \in \mathbb{C}$  and  $a, b \in \mathbb{R}$  with  $a - b \in \mathbb{Z}$ , set

$$\eta(x; a, b) = \prod_{r=1}^{b-a} \frac{[x + a + r]}{[-x + a + r]} \quad \text{if } b > a$$

$$\eta(x; a, b) = \prod_{r=1}^{a-b} \frac{[-x + b + r]}{[x + b + r]} \quad \text{if } a > b,$$

and  $\eta(x; a, b) = 1$  if  $a = b$ . Note the identities  $\eta(x; a, b) = \eta(x; b, a)^{-1}$  and

$$\eta(x; a, b) \eta(x; b, c) = \eta(x; a, c)$$

if  $a - b, b - c \in \mathbb{Z}$

Now we state one of the main theorems of this paper yielding the tetrahedral symmetry for the trigonometric  $6j$ -symbols. This theorem describes the behavior of the trigonometric  $6j$ -symbols under the permutation of the second and third columns. Together with the symmetries established in Theorem 6.3.1 this yields the transformation properties of the trigonometric  $6j$ -symbols with respect to the action of the symmetric group  $S_4$ .

**7.2. Theorem.** For any integers  $i, j, k, l, m, n \geq 0$  such that the triples  $(i, j, n)$ ,  $(i, l, m)$ ,  $(j, k, m)$ ,  $(k, l, n)$  are admissible and any  $x \in \mathbb{C}$ , we have

$$\left| \begin{smallmatrix} i & n & j \\ k & m & l \end{smallmatrix} \right| (x) = \frac{\pi_{j,l}(x-1)}{\pi_{m,n}(x-1)} \eta(x; \frac{m+n}{2}, \frac{j+l}{2}) \left| \begin{smallmatrix} i & j & n \\ k & l & m \end{smallmatrix} \right| (x). \quad (7.2a)$$

It is easy to deduce from the results of Section 6.3 that when  $x$  converges to  $+\infty$  formula (7.2a) degenerates into the equality

$$\left| \begin{smallmatrix} i & n & j \\ k & m & l \end{smallmatrix} \right| = \left| \begin{smallmatrix} i & j & n \\ k & l & m \end{smallmatrix} \right|.$$

Therefore formula (7.2a) may be regarded as a parametrized version of the tetrahedral symmetry for  $6j$ -symbols without spectral parameter.

The proof of Theorem 7.2 given below relies on a computation of the trigonometric  $6j$ -symbols in the case where one of the six entries is 0. Specifically, we shall use the following formula:

$$\left\{ \begin{smallmatrix} m & r & n \\ n & 0 & m \end{smallmatrix} \right\} (x) = \pi_{m,n}(x-1) \eta(x; \frac{r}{2}, \frac{m+n}{2}) \theta(m, n, r). \quad (7.2b)$$

This formula is proven in Section 8.

**Proof of Theorem.** We first establish the following formula expressing the trigonometric  $6j$ -symbols via the  $6j$ -symbols without spectral parameter:

$$\begin{aligned} \left| \begin{smallmatrix} i & j & n \\ k & l & m \end{smallmatrix} \right| (x) &= \pi_{m,n}(x-1) \sum_r \Delta(r) \eta(x; \frac{r}{2}, \frac{m+n}{2}) \left| \begin{smallmatrix} j & l & r \\ n & m & k \end{smallmatrix} \right| \left| \begin{smallmatrix} j & l & r \\ m & n & i \end{smallmatrix} \right|. \end{aligned} \quad (7.2c)$$

The same argument as in Lemma 6.7 shows the equality in Figure 7.1 where  $C_{m,n}^r(x) \in \mathbb{C}$ . The  $r$ -th summand on the right-hand side is the product of  $C_{m,n}^r(x)$  and  $a_r^{m,n} a_{m,n}^r$  where  $a_r^{m,n} \in TL_{r,m+n}$ ,  $a_{m,n}^r \in TL_{m+n,r}$  are the generators specified in Section 6.7. Multiplying both parts of the equality in Figure 7.1 by  $a_{r_0}^{m,n}$  from below and by  $a_{m,n}^{r_0}$  from above, taking the closure, and ap-

Figure 7.1

plying the bracket, we obtain

$$\left\{ \begin{matrix} m & r_0 & n \\ n & 0 & m \end{matrix} \right\} (x) = C_{m,n}^{r_0}(x) (\Delta(r_0))^{-1} (\theta(m, n, r_0))^2.$$

Hence for any  $r$ ,

$$\begin{aligned} C_{m,n}^r(x) &= \Delta(r) \left\{ \begin{matrix} m & r & n \\ n & 0 & m \end{matrix} \right\} (x) (\theta(m, n, r))^{-2} \\ &= \Delta(r) \pi_{m,n}(x-1) \eta(x; \frac{r}{2}, \frac{m+n}{2}) (\theta(m, n, r))^{-1}. \end{aligned}$$

The computation in Figure 7.2 (which uses the equalities in Figures 7.1, 6.6 and Lemma 6.6) shows that

$$\left\{ \begin{matrix} i & j & n \\ k & l & m \end{matrix} \right\} (x) = \sum_r \frac{C_{m,n}^r(x)}{\theta(r, j, l)} \left\{ \begin{matrix} j & l & r \\ n & m & k \end{matrix} \right\} \left\{ \begin{matrix} j & l & r \\ m & n & i \end{matrix} \right\}.$$

Substituting in the last expression the value of  $C_{m,n}^r(x)$  and dividing by the product  $\tilde{\theta}(i, j, n) \tilde{\theta}(i, l, m) \tilde{\theta}(j, k, m) \tilde{\theta}(k, l, n)$  we obtain (7.2.c).

By (7.2.c),

$$\begin{aligned} \left| \begin{matrix} i & n & j \\ k & m & l \end{matrix} \right| (x) &= \pi_{j,l}(x-1) \sum_r \Delta(r) \eta(x; \frac{r}{2}, \frac{j+l}{2}) \left| \begin{matrix} n & m & r \\ j & l & k \end{matrix} \right| \left| \begin{matrix} n & m & r \\ l & j & i \end{matrix} \right| \\ &= \frac{\pi_{j,l}(x-1)}{\pi_{m,n}(x-1)} \eta(x; \frac{m+n}{2}, \frac{j+l}{2}) \\ &\quad \times \pi_{m,n}(x-1) \sum_r \Delta(r) \eta(x; \frac{r}{2}, \frac{m+n}{2}) \left| \begin{matrix} j & l & r \\ n & m & k \end{matrix} \right| \left| \begin{matrix} j & l & r \\ m & n & i \end{matrix} \right| \\ &= \frac{\pi_{j,l}(x-1)}{\pi_{m,n}(x-1)} \eta(x; \frac{m+n}{2}, \frac{j+l}{2}) \left| \begin{matrix} i & j & n \\ k & l & m \end{matrix} \right| (x). \end{aligned}$$



$$\begin{aligned}
 & \begin{array}{c} k \\ \diagup \quad \diagdown \\ l \quad x \quad j \\ \diagdown \quad \diagup \\ i \end{array} \doteq \sum_r C_{m,n}^r(x) \begin{array}{c} k \\ \diagup \quad \diagdown \\ l \quad n \quad m \\ \diagdown \quad \diagup \\ i \end{array} j \doteq \\
 & \doteq \sum_r C_{m,n}^r(x) \frac{\Delta(l)}{\theta(r,j,l)} \begin{array}{c} k \\ \diagup \quad \diagdown \\ l \quad n \quad m \\ \diagdown \quad \diagup \\ i \end{array} j \doteq \\
 & \doteq \sum_r \frac{C_{m,n}^r(x)}{\theta(r,j,l)} \begin{array}{c} k \\ \diagup \quad \diagdown \\ l \quad n \quad m \\ \diagdown \quad \diagup \\ i \end{array} j \doteq
 \end{aligned}$$

Figure 7.2

**7.3. Corollaries.** The next three formulas describe the behavior of the trigonometric  $6j$ -symbol under permutations of the first column with the third and second ones and under the transformation  $x \mapsto 1 - x$ :

$$\left| \begin{array}{ccc} n & j & i \\ m & l & k \end{array} \right| (x) = \frac{\pi_{i,k}(x)}{\pi_{m,n}(x)} \eta(1-x; \frac{m+n}{2}, \frac{i+k}{2}) \left| \begin{array}{ccc} i & j & n \\ k & l & m \end{array} \right| (x), \quad (7.3a)$$

$$\begin{aligned}
 \left| \begin{array}{ccc} j & i & n \\ l & k & m \end{array} \right| (x) &= \frac{\pi_{j,l}(x-1)}{\pi_{j,l}(x)} \frac{\pi_{i,k}(x)}{\pi_{i,k}(x-1)} \\
 &\times \eta(x; \frac{i+k}{2}, \frac{j+l}{2}) \eta(1-x; \frac{j+l}{2}, \frac{i+k}{2}) \left| \begin{array}{ccc} i & j & n \\ k & l & m \end{array} \right| (x), \quad (7.3b)
 \end{aligned}$$

$$\begin{aligned}
 \left| \begin{array}{ccc} i & j & n \\ k & l & m \end{array} \right| (1-x) &= (-1)^{mn} \frac{\pi_{j,l}(x-1)}{\pi_{j,l}(x)} \frac{\pi_{i,k}(x)}{\pi_{i,k}(x-1)} \\
 &\times \eta(x; \frac{i+k}{2}, \frac{j+l}{2}) \eta(1-x; \frac{j+l}{2}, \frac{i+k}{2}) \left| \begin{array}{ccc} i & j & n \\ k & l & m \end{array} \right| (x). \quad (7.3c)
 \end{aligned}$$

Let us prove (7.3a). Using formulas (6.3.e), (7.2.a), once more (6.3.e), and finally (6.3.f), we obtain

$$\begin{aligned}
 \left| \begin{smallmatrix} i & j & n \\ k & l & m \end{smallmatrix} \right| (x) &= (-1)^{mn} \left| \begin{smallmatrix} j & k & m \\ l & i & n \end{smallmatrix} \right| (1-x) \\
 &= (-1)^{mn} \frac{\pi_{m,n}(-x)}{\pi_{i,k}(-x)} \eta\left(1-x; \frac{i+k}{2}, \frac{m+n}{2}\right) \left| \begin{smallmatrix} j & m & k \\ l & n & i \end{smallmatrix} \right| (1-x) \\
 &= (-1)^{mn+ik} \frac{\pi_{m,n}(-x)}{\pi_{i,k}(-x)} \eta\left(1-x; \frac{i+k}{2}, \frac{m+n}{2}\right) \left| \begin{smallmatrix} m & l & i \\ n & j & k \end{smallmatrix} \right| (x) \\
 &= \frac{\pi_{m,n}(x)}{\pi_{i,k}(x)} \eta\left(1-x; \frac{i+k}{2}, \frac{m+n}{2}\right) \left| \begin{smallmatrix} n & j & i \\ m & l & k \end{smallmatrix} \right| (x).
 \end{aligned}$$

The resulting equality is equivalent to (7.3a).

By (7.2a), (7.3a), and once more (7.2a),

$$\begin{aligned}
 &\left| \begin{smallmatrix} i & j & n \\ k & l & m \end{smallmatrix} \right| (x) \\
 &= \frac{\pi_{m,n}(x-1)}{\pi_{j,l}(x-1)} \eta\left(x; \frac{j+l}{2}, \frac{m+n}{2}\right) \left| \begin{smallmatrix} i & n & j \\ k & m & l \end{smallmatrix} \right| (x) \\
 &= \frac{\pi_{m,n}(x-1)}{\pi_{j,l}(x-1)} \eta\left(x; \frac{j+l}{2}, \frac{m+n}{2}\right) \\
 &\quad \times \frac{\pi_{j,l}(x)}{\pi_{i,k}(x)} \eta\left(1-x; \frac{i+k}{2}, \frac{j+l}{2}\right) \left| \begin{smallmatrix} j & n & i \\ l & m & k \end{smallmatrix} \right| (x) \\
 &= \frac{\pi_{m,n}(x-1) \pi_{j,l}(x) \pi_{i,k}(x-1)}{\pi_{j,l}(x-1) \pi_{i,k}(x) \pi_{m,n}(x-1)} \\
 &\quad \times \eta\left(x; \frac{j+l}{2}, \frac{m+n}{2}\right) \eta\left(1-x; \frac{i+k}{2}, \frac{j+l}{2}\right) \eta\left(x; \frac{m+n}{2}, \frac{i+k}{2}\right) \left| \begin{smallmatrix} j & i & n \\ l & k & m \end{smallmatrix} \right| (x) \\
 &= \frac{\pi_{j,l}(x) \pi_{i,k}(x-1)}{\pi_{j,l}(x-1) \pi_{i,k}(x)} \eta\left(x; \frac{j+l}{2}, \frac{i+k}{2}\right) \eta\left(1-x; \frac{i+k}{2}, \frac{j+l}{2}\right) \left| \begin{smallmatrix} j & i & n \\ l & k & m \end{smallmatrix} \right| (x).
 \end{aligned}$$

The resulting equality is equivalent to (7.3b).

By (6.3e), (6.3g), and (7.3b),

$$\begin{aligned}
 (-1)^{mn} \left| \begin{smallmatrix} i & j & n \\ k & l & m \end{smallmatrix} \right| (x) &= \left| \begin{smallmatrix} j & k & m \\ l & i & n \end{smallmatrix} \right| (1-x) = \left| \begin{smallmatrix} j & i & n \\ l & k & m \end{smallmatrix} \right| (1-x) \\
 &= \frac{\pi_{j,l}(x) \pi_{i,k}(x-1)}{\pi_{j,l}(x-1) \pi_{i,k}(x)} \eta\left(1-x; \frac{i+k}{2}, \frac{j+l}{2}\right) \eta\left(x; \frac{j+l}{2}, \frac{i+k}{2}\right) \left| \begin{smallmatrix} i & j & n \\ k & l & m \end{smallmatrix} \right| (1-x).
 \end{aligned}$$

The resulting equality is equivalent to (7.3c).

#### 7.4. Renormalized trigonometric $6j$ -symbols

We can renormalize trigonometric  $6j$ -symbols simplifying their behavior under the tetrahedral symmetries. Set

$$\left| \begin{array}{ccc} i & j & n \\ k & l & m \end{array} \right|' (x) = \frac{(-1)^{(m-i-l)/2}}{\pi_{m,n-1}(x - \frac{1}{2})} \prod_{r=\frac{i-k}{2}}^{\frac{i+l}{2}} [x+r] \prod_{r=-\frac{i+k}{2}-1}^{\frac{i-l}{2}-1} [x+r] \left| \begin{array}{ccc} i & j & n \\ k & l & m \end{array} \right| (x).$$

Note that

$$\pi_{m,n-1}(x - \frac{1}{2}) = \pi_{m,n}(x) \prod_{r=\frac{n-m}{2}}^{\frac{m+n}{2}-1} [x+r]^{-1}.$$

This modified  $6j$ -symbol admits an interesting geometric interpretation which will be discussed in Section 8.10. Observe also that this  $6j$ -symbol is invariant under the transformation  $x \mapsto 1-x$  and under 6 permutations generated by the following two:

$$\left| \begin{array}{ccc} i & j & n \\ k & l & m \end{array} \right|' (x) = \left| \begin{array}{ccc} l & k & n \\ j & i & m \end{array} \right|' (x) = \left| \begin{array}{ccc} n & j & i \\ m & l & k \end{array} \right|' (x). \quad (7.4a)$$

These facts may be deduced from the symmetries established above and (7.3.c) or from the explicit formula for this  $6j$ -symbol given in Section 8.10.

There is another interesting renormalization of trigonometric  $6j$ -symbols. Set

$$\left| \begin{array}{ccc} i & j & n \\ k & l & m \end{array} \right|'' (x) = (-1)^{(m-i-l)/2} \prod_{r=-\frac{|l-j|}{2}}^{\frac{|l-j|}{2}-1} [x+r]^{\text{sign}(l-j)} \left| \begin{array}{ccc} i & j & n \\ k & l & m \end{array} \right|' (x)$$

where  $\text{sign}(s) = 1$  if  $s \geq 0$ ,  $\text{sign}(s) = -1$  if  $s < 0$ . It can be deduced from the symmetries established above that this renormalized  $6j$ -symbol has a full tetrahedral symmetry, i.e., it is preserved under permutations of columns in the matrix

$$\begin{bmatrix} i & j & n \\ k & l & m \end{bmatrix}$$

as well as under the permutation of rows in any  $(2 \times 2)$ -submatrix.

**7.5. Proof of (4.3b).** It suffices to consider the case where  $L$  is a vertical interval colored with  $m$  and  $L_1$  is obtained from  $L$  by  $\omega_1$  as in Figure 1.1. By

Lemma 6.6, it suffices to show that the bracket function of the closure of  $L_1$  is equal to

$$(-1)^m \frac{[x - m - 1]}{[x - 1]} \pi_{m,m}(x - 1) \Delta(m). \quad (7.5a)$$

It follows from definitions, (4.3a), and (7.2b) that this bracket function equals

$$\begin{aligned} \left\{ \begin{matrix} 0 & m & m \\ 0 & m & m \end{matrix} \right\} (x) &= (-1)^m \left\{ \begin{matrix} m & 0 & m \\ m & 0 & m \end{matrix} \right\} (1 - x) \\ &= (-1)^m \pi_{m,m}(-x) \eta(1 - x; 0, m) \theta(m, m, 0) \\ &= \pi_{m,m}(x) \eta(1 - x; 0, m) \Delta(m). \end{aligned}$$

It is easy to check that the last expression equals (7.5a).

## 8. Computation of the trigonometric 6j-symbols

In this section we explicitly compute the trigonometric 6j-symbols and in particular prove formula (7.2.b).

**8.1. Notation.** Let  $i, j, k, l, m, n$  be nonnegative integers such that the triples  $(i, j, n), (i, l, m), (j, k, m), (k, l, n)$  are admissible. For  $x \in \mathbb{C}$ , set

$$v_{m,n}(i, j, k, l | x) = \frac{\left\{ \begin{matrix} i & j & n \\ k & l & m \end{matrix} \right\} (x)}{\left\{ \begin{matrix} i & j & n \\ k & l & m \end{matrix} \right\}} = \frac{\left| \begin{matrix} i & j & n \\ k & l & m \end{matrix} \right| (x)}{\left| \begin{matrix} i & j & n \\ k & l & m \end{matrix} \right|} \in \mathbb{C}.$$

Note that by (6.3.b) and (6.3.c),

$$v_{m,0}(i, j, k, l | x) = 1, \quad v_{0,n}(i, j, k, l | x) = 1.$$

We shall be mainly concerned with computing  $v_{m,n}(i, j, k, l | x)$ . Using the well known formulas for the 6j-symbols without spectral parameter we can always recover the numbers

$$\left\{ \begin{matrix} i & j & n \\ k & l & m \end{matrix} \right\} (x) = \left\{ \begin{matrix} i & j & n \\ k & l & m \end{matrix} \right\} \cdot v_{m,n}(i, j, k, l | x)$$

and

$$\left| \begin{matrix} i & j & n \\ k & l & m \end{matrix} \right| (x) = \left| \begin{matrix} i & j & n \\ k & l & m \end{matrix} \right| \cdot v_{m,n}(i, j, k, l | x).$$

**8.2. Case  $m = n = 1$ .** In the case  $m = n = 1$  the admissibility of the triples  $(i, j, n), (i, l, m), (j, k, m), (k, l, n)$  implies that there are only three cases to consider:

- (i)  $i = l \pm 1, j = l \pm 2, k = i = l \pm 1$ , or
- (ii)  $i = l \pm 1, j = l, k = i = l \pm 1$ , or
- (iii)  $i = l \pm 1, j = l, k = l \mp 1$ .

We claim that

$$v_{1,1}(l \pm 1, l \pm 2, l \pm 1, l | x) = [x - 1], \quad (8.2a)$$

$$v_{1,1}(l \pm 1, l, l \pm 1, l | x) = -[x \pm (l + 1)], \quad (8.2b)$$

$$v_{1,1}(l \pm 1, l, l \mp 1, l | x) = [x]. \quad (8.2c)$$

To prove these formulas, consider the element  $R(x) = [x]e_1 + [x - 1]1_2$  of the Temperley-Lieb algebra  $TL_2$  (cf. Section 4.5). Recall Kauffman's  $R$ -matrix  $R_+ = q^{-1/2}e_1 + q^{1/2}1_2 \in TL_2$ . Set  $R_- = R_+^{-1} = q^{1/2}e_1 + q^{-1/2}1_2$ . A direct computation shows that

$$R(x) = \frac{q^{x-1/2}R_- - q^{-x+1/2}R_+}{q - q^{-1}}.$$

Substituting this formula in the definition of

$$\left\{ \begin{matrix} i & j & 1 \\ k & l & 1 \end{matrix} \right\} (x)$$

we obtain the equality in Figure 8.1. The right-hand side can be computed using the formulas in Figure 6.3 where  $q_a = \sqrt{-1}^a q^{a(a+2)/4}$  for an integer  $a$ . This gives

$$\left\{ \begin{matrix} i & j & 1 \\ k & l & 1 \end{matrix} \right\} (x) = \frac{q^{x-1/2}q_i q_j^{-1} q_k q_l^{-1} - q^{-x+1/2}q_i^{-1} q_j q_k^{-1} q_l}{q - q^{-1}} \left\{ \begin{matrix} i & j & 1 \\ k & l & 1 \end{matrix} \right\}.$$

Now, a case by case computation gives equalities (8.2.a) – (8.2.c).

**Figure 8.1**

**8.3. Lemma.** *Let  $i, j, k, l, m, n_1, n_2, p$  be nonnegative integers such that the triples  $(i, j, n_1 + n_2), (i, l, m), (j, k, m), (k, p, n_2), (l, p, n_1)$  are admissible. If there is only one integer  $r \geq 0$  such that the triples  $(r, p, m), (r, n_1, i), (r, n_2, j)$  are admissible then*

$$v_{m, n_1 + n_2}(i, j, k, l | x) = v_{m, n_1}(i, r, p, l | x_1) v_{m, n_2}(r, j, k, p | x_2) \quad (8.3a)$$

where  $x_1 = x - n_2/2, x_2 = x + n_1/2$ .

This lemma and its versions stated below are the key to the inductive computation of the trigonometric  $6j$ -symbols.

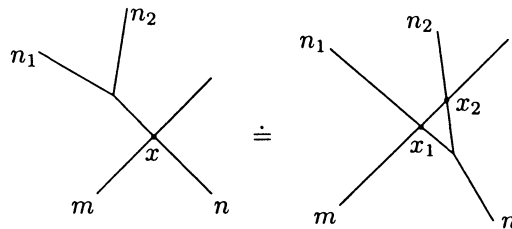
**Proof of Lemma.** Set  $n = n_1 + n_2$ . Consider first the case  $n_1 \neq 0, n_2 \neq 0$ . The key observation is the equality of brackets in Figure 8.2. This equality follows from definitions and the formula

$$(F_n \otimes F_m)X_{m,n}(x)(F_m \otimes F_n) \doteq (1_n \otimes F_m)T_{m,n}(x)(F_m \otimes F_n)$$

established in Section 5. The equality in Figure 8.2 implies the formula in Figure 8.3 (where the dotted lines should be ignored for the moment). Note that the admissibility of the triples  $(i, j, n), (i, l, m), (j, k, m), (k, p, n_2), (l, p, n_1)$  guarantees our right to consider the colored graph-nets in Figure 8.3. Applying Lemma 6.7 to the edges connected by dotted lines we can compute the brackets of these graphs. This gives the equality

$$\begin{aligned} & (\theta(k, l, n))^{-1} \left\{ \begin{matrix} n_1 & n_2 & n \\ k & l & p \end{matrix} \right\} \left\{ \begin{matrix} i & j & n \\ k & l & m \end{matrix} \right\} (x) \\ &= \sum_r \Delta(r) \frac{\left\{ \begin{matrix} i & r & n_1 \\ p & l & m \end{matrix} \right\} (x_1) \left\{ \begin{matrix} r & j & n_2 \\ k & p & m \end{matrix} \right\} (x_2) \left\{ \begin{matrix} n_1 & n_2 & n \\ j & i & r \end{matrix} \right\}}{\theta(r, p, m) \theta(r, n_1, i) \theta(r, n_2, j)}. \end{aligned} \quad (8.3b)$$

On the right-hand side we sum over all  $r$  such that the triples  $(r, p, m), (r, n_1, i), (r, n_2, j)$  are admissible. Since there is only one  $r$ , the sum has 1 summand.



**Figure 8.2**

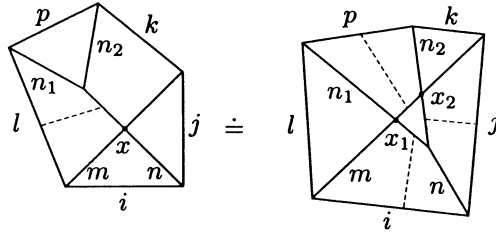


Figure 8.3

Similar computations apply to  $6j$ -symbols without spectral parameter, it suffices to replace crossings with overcrossings as in Figure 4.4 and to use the first equality in Figure 6.3. This gives

$$\begin{aligned}
 & (\theta(k, l, n))^{-1} \left\{ \begin{matrix} n_1 & n_2 & n \\ k & l & p \end{matrix} \right\} \left\{ \begin{matrix} i & j & n \\ k & l & m \end{matrix} \right\} \\
 &= \sum_r \Delta(r) \frac{\left\{ \begin{matrix} i & r & n_1 \\ p & l & m \end{matrix} \right\} \left\{ \begin{matrix} r & j & n_2 \\ k & p & m \end{matrix} \right\} \left\{ \begin{matrix} n_1 & n_2 & n \\ j & i & r \end{matrix} \right\}}{\theta(r, p, m) \theta(r, n_1, i) \theta(r, n_2, j)}. \quad (8.3.c)
 \end{aligned}$$

By assumption, the sum on the right-hand side has only one summand corresponding to the same  $r$  as above. It follows from (6.2.c) that

$$\left\{ \begin{matrix} n_1 & n_2 & n \\ k & l & p \end{matrix} \right\} \neq 0 \quad \text{and} \quad \left\{ \begin{matrix} n_1 & n_2 & n \\ j & i & r \end{matrix} \right\} \neq 0.$$

Dividing (8.3.b) by (8.3.c) we obtain (8.3.a).

In the case  $n_1 = 0$  or  $n_2 = 0$  equality (8.3.a) holds by obvious reasons. Assume for concreteness that  $n_1 = 0$ . Then  $x_2 = x$  and  $v_{m,n_1}(i, r, p, l | x_1) = 1$ . The admissibility of the triples  $(l, p, n_1)$  and  $(r, n_1, i)$  implies that  $p = l$  and  $r = i$ . Thus, formula (8.3.a) becomes the identity  $v_{m,n_1}(i, j, k, l | x) = v_{m,n_1}(i, j, k, l | x)$ .

**8.4. Lemma.** *Let  $i, j, k, l, m, n_1, n_2, p$  be nonnegative integers such that the triples  $(i, l, m)$ ,  $(i, p, n_1)$ ,  $(j, p, n_2)$ ,  $(j, k, m)$ ,  $(k, l, n_1 + n_2)$  are admissible. If there is only one integer  $r \geq 0$  such that the triples  $(r, p, m)$ ,  $(r, n_1, l)$ ,  $(r, n_2, k)$  are admissible then*

$$v_{m,n_1+n_2}(i, j, k, l | x) = v_{m,n_1}(i, p, r, l | x_1) v_{m,n_2}(p, j, k, r | x_2)$$

where  $x_1 = x + n_2/2$ ,  $x_2 = x - n_1/2$ .

The proof of this lemma is analogous to the proof of Lemma 8.3, instead of the equality in Figure 8.2 one should use the equality in Figure 8.4. (Al-

ternatively, one can deduce Lemma 8.4 from Lemma 8.3 using symmetries of  $6j$ -symbols.)

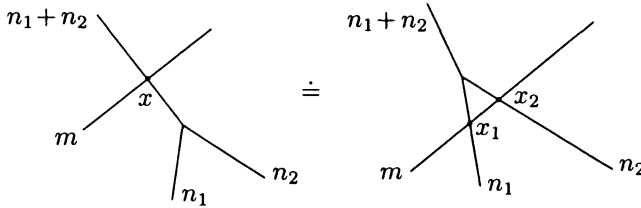


Figure 8.4

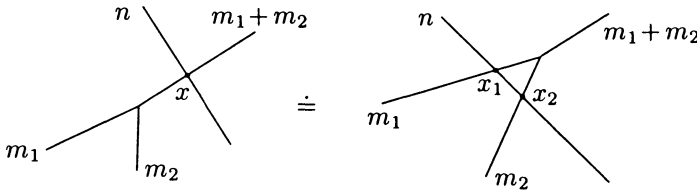


Figure 8.5

In Section 8.7 we shall need the following version of Lemma 8.3. Its proof is based on the equality in Figure 8.5.

**8.5. Lemma.** *Let  $i, j, k, l, m_1, m_2, n, p$  be nonnegative integers such that the triples  $(i, j, n), (i, p, m_2), (j, k, m_1 + m_2), (k, l, n), (l, p, m_1)$  are admissible. If there is only one integer  $r \geq 0$  such that the triples  $(r, p, n), (r, m_1, k), (r, m_2, j)$  are admissible then*

$$v_{m_1+m_2, n}(i, j, k, l | x) = v_{m_1, n}(p, r, k, l | x_1) v_{m_2, n}(i, j, r, p | x_2)$$

where  $x_1 = x - m_2/2, x_2 = x + m_1/2$ .

**8.6. Case  $m = 1, n \geq 1$ .** For  $x \in \mathbb{C}$  and an integer  $n \geq 1$ , set

$$\{x\}_{n-1} = \prod_{r=-\frac{n-1}{2}}^{\frac{n-1}{2}-1} [x + r].$$

In this subsection we compute  $v_{m, n}(i, j, k, l | x)$  with  $m = 1, n \geq 1$ . The admissibility of the triples  $(i, j, n), (i, l, m), (j, k, m), (k, l, n)$  implies that  $i = l + \varepsilon$



and  $j = k + \varepsilon'$  where  $\varepsilon, \varepsilon' = \pm 1$ . We claim that

$$v_{1,n}(l+1, k \pm 1, k, l | x) = \pm [x + \frac{l \mp k \mp 1}{2}] \{x\}_{n-1}, \quad (8.6a)$$

$$v_{1,n}(l-1, k \pm 1, k, l | x) = \mp [x \mp \frac{(k+1) \pm (l+1) \pm 1}{2}] \{x\}_{n-1}. \quad (8.6b)$$

The remaining part of Section 8.6 is devoted to the proof of (8.6a), (8.6b). One has to consider four cases depending on  $\varepsilon = i - l, \varepsilon' = j - k$ . Consider first the case  $\varepsilon = \varepsilon' = +1$ . We shall use two inductive formulas which are special cases of Lemmas 8.3 and 8.4. Take in Lemma 8.3

$$m = 1, n_1 = n - 1, n_2 = 1, i = l + 1, j = k + 1, p = k - 1.$$

Assume that  $k + l + n = 0 \pmod{2}$  and

$$|k - l - 1| + 1 \leq n \leq k + l + 2.$$

Then the conditions of Lemma 8.3 are fulfilled, the only integer  $r \geq 0$  such that the triples  $(r, p, m), (r, n_1, i), (r, n_2, j)$  are admissible equals  $k$ . By Lemma 8.3,

$$\begin{aligned} v_{1,n}(l+1, k+1, k, l | x) \\ = v_{1,n-1}(l+1, k, k-1, l | x - 1/2) v_{1,1}(k, k+1, k, k-1 | x + (n-1)/2). \end{aligned}$$

By (8.2a), this yields

$$v_{1,n}(l+1, k+1, k, l | x) = [x + \frac{n-1}{2} - 1] v_{1,n-1}(l+1, k, k-1, l | x - 1/2). \quad (8.6c)$$

Similarly, take in Lemma 8.4

$$m = 1, n_1 = n - 1, n_2 = 1, i = l + 1, j = k + 1, p = k + 2.$$

Assume that  $k + l + n = 0 \pmod{2}$  and

$$|l - k - 1| + 1 \leq n \leq k + l + 2.$$

It is clear that the triples  $(r, p, m), (r, n_1, i), (r, n_2, j)$  are admissible only for  $r = k + 1$ . By Lemma 8.4,

$$\begin{aligned} v_{1,n}(l+1, k+1, k, l | x) \\ = v_{1,n-1}(l+1, k+2, k+1, l | x + 1/2) v_{1,1}(k+2, k+1, k, k+1 | x - (n-1)/2). \end{aligned}$$

By (8.2c), this yields

$$v_{1,n}(l+1, k+1, k, l | x) = [x - (n-1)/2] v_{1,n-1}(l+1, k+2, k+1, l | x+1/2). \quad (8.6d)$$

Now, we can compute  $v_{1,n}(l+1, k+1, k, l | x)$ . Set  $h = (n + l - k)/2$ . Applying (8.6d) inductively  $h$  times we obtain

$$v_{1,n}(l+1, k+1, k, l | x) = \prod_{t=0}^{h-1} [x - \frac{n-1}{2} + t] v_{1,n-h}(l+1, k+h+1, k+h, l | x + \frac{h}{2}).$$

Applying (8.6c) inductively  $n - h$  times we obtain

$$\begin{aligned} & v_{1,n-h}(l+1, k+h+1, k+h, l | x + \frac{h}{2}) \\ &= v_{1,0}(l+1, l+1, l, l | x + \frac{l-k}{2}) \prod_{u=1}^{n-h} [x + \frac{n-1}{2} - u] = \prod_{u=1}^{n-h} [x + \frac{n-1}{2} - u]. \end{aligned}$$

Combining these formulas we obtain

$$v_{1,n}(l+1, k+1, k, l | x) = [x + \frac{l-k-1}{2}] \{x\}_{n-1}.$$

In the case  $\varepsilon = -1$  and/or  $\varepsilon' = -1$  the computation is quite similar. Here are the relevant generalizations of (8.6c) and (8.6d):

$$\begin{aligned} v_{1,n}(l+\varepsilon, k+\varepsilon', k, l | x) &= [x + (n-1)/2 - 1] v_{1,n-1}(l+\varepsilon, k, k-\varepsilon', l | x-1/2), \\ v_{1,n}(l+\varepsilon, k+\varepsilon', k, l | x) &= [x - (n-1)/2] v_{1,n-1}(l+\varepsilon, k+2\varepsilon', k+\varepsilon', l | x+1/2). \end{aligned}$$

To establish these formulas one proceeds as above setting  $p = k - \varepsilon', r = k$  and  $p = k + 2\varepsilon', r = k + \varepsilon'$ , respectively.

### 8.7. Case $k = j+m$ .

In this subsection we compute  $v_{m,n}(i, j, k, l | x)$  in the case  $k = j+m$ . We claim that

$$\begin{aligned} & v_{m,n}(i, j, j+m, l | x) \tag{8.7a} \\ &= (-1)^{(i+m-l)/2} \pi_{m,n-1}(x - \frac{1}{2}) \prod_{r=1}^{\frac{l+m-i}{2}} [x + \frac{m+j-i}{2} - r] \\ &\quad \prod_{r=1}^{\frac{i+m-l}{2}} [x + \frac{m+i+j}{2} + 1 - r]. \end{aligned}$$

Substituting  $j = 0, i = n$ , we obtain

$$\begin{aligned} v_{m,n}(n, 0, m, l | x) \\ &= (-1)^{(n+m-l)/2} \pi_{m,n-1}(x - \frac{1}{2}) \prod_{r=1}^{\frac{l+m-n}{2}} [x + \frac{m-n}{2} - r] \prod_{r=1}^{\frac{n+m-l}{2}} [x + \frac{m+n}{2} + 1 - r] \\ &= \eta(x; \frac{l}{2}, \frac{m+n}{2}) \pi_{m,n}(x-1). \end{aligned}$$

This implies

$$\left\{ \begin{matrix} n & 0 & n \\ m & l & m \end{matrix} \right\} (x) = \eta(x; \frac{l}{2}, \frac{m+n}{2}) \pi_{m,n}(x-1) \theta(m, n, l).$$

The last formula implies (7.2b):

$$\left\{ \begin{matrix} m & r & n \\ n & 0 & m \end{matrix} \right\} (x) = \left\{ \begin{matrix} n & 0 & n \\ m & r & m \end{matrix} \right\} (x) = \eta(x; \frac{r}{2}, \frac{m+n}{2}) \pi_{m,n}(x-1) \theta(m, n, r).$$

To prove (8.7.a) we use induction similar to the one used in Section 8.6. We apply Lemma 8.5 to

$$m_1 = m - 1, m_2 = 1, k = j + m, p = i \pm 1.$$

We assume that the triples  $(i, j, n), (j + m, l, n), (j + m, p, m - 1), (j + 1, p, n)$  are admissible. There is only one integer  $r$  such that the triples  $(r, m - 1, j + m), (r, 1, j)$  are admissible, namely  $r = j + 1$ . By Lemma 8.5,

$$\begin{aligned} v_{m,n}(i, j, j + m, l | x) \\ &= v_{m-1,n}(p, j + 1, j + m, l | x - 1/2) v_{1,n}(i, j, j + 1, p | x + (m - 1)/2). \end{aligned} \tag{8.7.b}$$

Substituting here the value of  $v_{1,n}(i, j, j + 1, p | x + (m - 1)/2)$  computed in Section 8.6, we obtain for  $p = j + 1$ ,

$$\begin{aligned} v_{m,n}(i, j, j + m, l | x) \\ &= [x + \frac{m + j - i}{2} - 1] \{x + \frac{m - 1}{2}\}_{n-1} v_{m-1,n}(i + 1, j + 1, j + m, l | x - 1/2) \end{aligned} \tag{8.7c}$$

and for  $p = j - 1$ ,

$$\begin{aligned} v_{m,n}(i, j, j + m, l | x) \\ &= -[x + \frac{m + i + j}{2}] \{x + \frac{m - 1}{2}\}_{n-1} v_{m-1,n}(i - 1, j + 1, j + m, l | x - 1/2). \end{aligned} \tag{8.7d}$$

Using these two formulas we can compute  $v_{m,n}(i, j, j+m, l | x)$ . Applying (8.7c) consecutively  $h = (l + m - i)/2$  times we obtain

$$\begin{aligned} v_{m,n}(i, j, j+m, l | x) & \quad (8.7e) \\ &= v_{m-h,n}(i+h, j+h, j+m, l | x - \frac{h}{2}) \\ &= \prod_{r=1}^h [x + \frac{m+j-i}{2} - r] \{x + \frac{m+1}{2} - r\}_{n-1}. \end{aligned}$$

Applying (8.7d) consecutively  $m-h = (i+m-l)/2$  times and using the equality  $v_{0,n}(l, j+m, j+m, l | x - m/2) = 1$  we obtain

$$\begin{aligned} & v_{m-h,n}(i+h, j+h, j+m, l | x - h/2) \\ &= (-1)^{m-h} \prod_{r=1}^{m-h} [x + \frac{m+i+j}{2} + 1 - r] \{x + \frac{m+1}{2} - h - r\}_{n-1}. \end{aligned}$$

Substituting this expression in (8.7e) and using the equality

$$\prod_{r=1}^m \{x + \frac{m+1}{2} - r\}_{n-1} = \pi_{m,n-1}(x - \frac{1}{2})$$

we obtain (8.7a).

**8.8. Case  $j = k + m$**  Using (6.3g), (6.3e), and similar identities for  $6j$ -symbols without spectral parameter, we obtain

$$v_{m,n}(i, j, k, l | x) = v_{n,m}(i, l, k, j | x) = (-1)^{mn} v_{m,n}(l, k, j, i | 1 - x).$$

By (8.7a),

$$\begin{aligned} v_{m,n}(i, k+m, k, l | x) &= (-1)^{mn} v_{m,n}(l, k, k+m, i | 1-x) \\ &= (-1)^{\frac{l+m-i}{2}} \pi_{m,n-1}(x - \frac{1}{2}) \prod_{r=1}^{\frac{i+m-l}{2}} [x - 1 - \frac{m+k-l}{2} + r] \\ &\quad \prod_{r=1}^{\frac{l+m-i}{2}} [x - 2 - \frac{m+k+l}{2} + r]. \end{aligned} \quad (8.8a)$$

### 8.9. General case

Now we can give an explicit formula for

$$\left\{ \begin{matrix} i & j & n \\ k & l & m \end{matrix} \right\} (x) \quad (8.9a)$$

where  $x \in \mathbb{C}$  and  $i, j, k, l, m, n$  are non-negative integers such that the triples  $(i, j, n)$ ,  $(i, l, m)$ ,  $(j, k, m)$ ,  $(k, l, n)$  are admissible.

Set

$$\alpha = \frac{j + m - k}{2}, \quad \beta = \frac{k + m - j}{2}, \quad \gamma = \frac{j + k - m}{2}.$$

The same argument as in the proof of Lemma 8.3 (using the equality in Figure 8.6 where  $x_1 = x + \alpha/2$ ,  $x_2 = x - \beta/2$ ) yields the identity

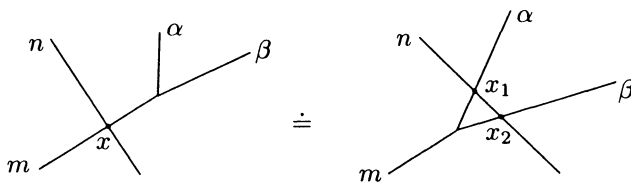
$$\left\{ \begin{matrix} i & j & n \\ k & l & m \end{matrix} \right\} (x) = \sum_r \Delta(r) \frac{\left\{ \begin{matrix} r & \gamma & n \\ k & l & \beta \end{matrix} \right\} (x_1) \left\{ \begin{matrix} i & j & n \\ \gamma & r & \alpha \end{matrix} \right\} (x_2) \left\{ \begin{matrix} i & \alpha & r \\ \beta & l & m \end{matrix} \right\}}{\theta(r, \beta, l) \theta(r, \gamma, n) \theta(r, \alpha, i)}. \quad (8.9b)$$

Here the summation goes over such  $r$  that the three  $6j$ -symbols on the right-hand side are well defined. We have

$$\left\{ \begin{matrix} r & \gamma & n \\ k & l & \beta \end{matrix} \right\} (x_1) = \left\{ \begin{matrix} r & \gamma & n \\ k & l & \beta \end{matrix} \right\} \cdot v_{\beta, n}(r, \gamma, k, l | x_1)$$

and

$$\left\{ \begin{matrix} i & j & n \\ \gamma & r & \alpha \end{matrix} \right\} (x_2) = \left\{ \begin{matrix} i & j & n \\ \gamma & r & \alpha \end{matrix} \right\} \cdot v_{\alpha, n}(i, j, \gamma, r | x_2).$$



**Figure 8.6**

Observe that the  $6j$ -symbols

$$\left\{ \begin{matrix} i & \alpha & r \\ \beta & l & m \end{matrix} \right\}, \quad \left\{ \begin{matrix} r & \gamma & n \\ k & l & \beta \end{matrix} \right\}, \quad \left\{ \begin{matrix} i & j & n \\ \gamma & r & \alpha \end{matrix} \right\}$$

are monomials which can be computed using (6.2c) and the tetrahedral symmetry of  $6j$ -symbols without spectral parameter. Substituting the expressions for

$$v_{\beta,n}(r, \gamma, k, l | x_1) = v_{\beta,n}(r, \gamma, \beta + \gamma, l | x_1),$$

$$v_{\alpha,n}(i, j, \gamma, r | x_2) = v_{\alpha,n}(i, \alpha + \gamma, \gamma, r | x_2)$$

obtained in Sections 8.7, 8.8 we get an explicit formula for (8.9a). We rewrite this formula as follows.

**8.10. Theorem.** *Set  $\sigma = \alpha + \beta + \gamma = (j + k + m)/2$  and*

$$\begin{aligned} & \left\{ \begin{matrix} i & j & n \\ k & l & m \end{matrix} \right\}' (x) \\ &= \sum_r (-1)^r \Delta(r) \left\{ \begin{matrix} i & \alpha & r \\ \beta & l & m \end{matrix} \right\} \left\{ \begin{matrix} l & \beta & r \\ \gamma & n & k \end{matrix} \right\} \left\{ \begin{matrix} n & \gamma & r \\ \alpha & i & j \end{matrix} \right\} \\ & \times (\theta(r, \beta, l) \theta(r, \gamma, n) \theta(r, \alpha, i))^{-1} \prod_{s=0}^{\sigma} \left[ x + \frac{\sigma - r}{2} - s - 1 \right] \left[ x + \frac{\sigma + r}{2} - s \right]. \end{aligned} \quad (8.10a)$$

Then

$$\begin{aligned} & \left\{ \begin{matrix} i & j & n \\ k & l & m \end{matrix} \right\} (x) = \\ & (-1)^{\frac{m-i-l}{2}} \pi_{m,n-1} \left( x - \frac{1}{2} \right) \prod_{r=\frac{i-k}{2}}^{\frac{i+l}{2}} [x + r]^{-1} \prod_{r=-\frac{i+k}{2}-1}^{\frac{i-l}{2}-1} [x + r]^{-1} \left\{ \begin{matrix} i & j & n \\ k & l & m \end{matrix} \right\}' (x). \end{aligned} \quad (8.10b)$$

Renormalizing the  $6j$ -symbol (8.10a) in the same way as in Sections 6.2, 6.3 we obtain precisely the  $6j$ -symbol considered at the beginning of Section 7.4. Indeed, it follows from (8.10.b) that

$$\left| \begin{matrix} i & j & n \\ k & l & m \end{matrix} \right| (x) = \frac{\left\{ \begin{matrix} i & j & n \\ k & l & m \end{matrix} \right\}' (x)}{\bar{\theta}(i, j, n) \bar{\theta}(i, l, m) \bar{\theta}(j, k, m) \bar{\theta}(k, l, n)}. \quad (8.10c)$$

It is natural to associate with the  $6j$ -symbol (8.10a) the geometric picture consisting of a triangle whose barycenter is endowed with  $x \in \mathbb{C}$  and connected to the vertices; the 6 edges of the resulting graph are labelled with  $i, j, k, l, m, n$

as in Figure 8.7. Note that the symmetries in (7.4a) are induced by the automorphisms of this graph preserving the barycenter. A graphical interpretation of formula (8.1a) is given in Figure 8.8 where

$$Y_r = (-1)^r \Delta(r) (\theta(r, \beta, l) \theta(r, \gamma, n) \theta(r, \alpha, i))^{-1} \times \prod_{s=0}^{\sigma} [x + \frac{\sigma - r}{2} - s - 1][x + \frac{\sigma + r}{2} - s].$$

These observations suggest to extend the theory of nets to include nets with trivalent vertices endowed with complex numbers. The corresponding extension of the state sum model of Section 6.10 should assign to every trivalent vertex the  $6j$ -symbol (8.10c).

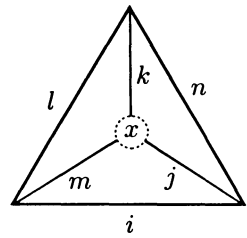


Figure 8.7

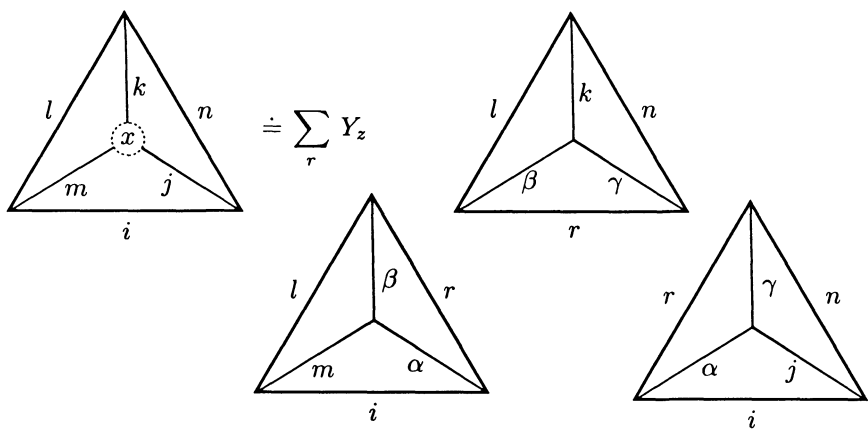


Figure 8.8

## 9. Trigonometric $6j$ -symbols as basic hypergeometric functions

### 9.1. Very-well-poised hypergeometric functions

We recall the definition of basic hypergeometric functions, see [GR]. Set

$$\begin{aligned} {}_{r+1}\phi_r(a_1, \dots, a_{r+1}; b_1, \dots, b_r; q, z) \\ = \sum_{n=0}^{\infty} \frac{(a_1; q)_n \cdots (a_{r+1}; q)_n}{(q; q)_n (b_1; q)_n \cdots (b_r; q)_n} z^n, \end{aligned} \quad (9.1a)$$

where

$$(a; q)_n = \begin{cases} 1 & , n = 0 \\ (1-a)(1-aq) \cdots (1-aq^{n-1}) & , n = 1, 2, \dots, \end{cases}$$

and  $a_1, \dots, a_{r+1}, b_1, \dots, b_r, q, z \in \mathbb{C}$ . An  ${}_{r+1}\phi_r$  series terminates if one of its numerator parameters is of the form  $q^{-m}$  with  $m = 0, 1, 2, \dots$  and  $q \neq 0$ . Since we will consider only the terminating series the question about convergence does not arise. More generally, one can define the  ${}_s\phi_r$  series though we will be interested only in the  $s = r + 1$  case.

An  ${}_{r+1}\phi_r$  series is called very-well-poised if

$$qa_1 = a_2b_1 = a_3b_2 = \cdots = a_{r+1}b_r, \quad a_2 = qa_1^{1/2}, \quad a_3 = -qa_1^{1/2} \quad (9.1b)$$

where we fixed a square root  $a_1^{1/2}$ . Following [GR] we use a more compact notation

$${}_{r+1}W_r(a_1; a_4, a_5, \dots, a_{r+1}; q, z) \quad (9.1c)$$

instead of  ${}_{r+1}\phi_r(a_1, \dots, a_{r+1}; b_1, \dots, b_r; q, z)$  for very-well-poised basic hypergeometric series. Note that the remaining parameters  $a_2, a_3, b_1, \dots, b_r$  are determined from (9.1b) if we assume that  $a_2 \neq 0, \dots, a_{r+1} \neq 0$ . Note also that the function (9.1c) is invariant under permutations of the parameters  $a_4, a_5, \dots, a_{r+1}$ , while  $a_1$  is distinguished. It will sometimes be convenient to rewrite  ${}_{r+1}W_r$  using the  $q$ -number notation  $[x] = \frac{q^x - q^{-x}}{q - q^{-1}}$  assuming as in the previous sections that  $q \neq 0$ .

The next lemma gives a convenient expression for the very-well-poised basic hypergeometric series (9.1c). We need the following notation: for  $\alpha \in \mathbb{C}$  and a natural number  $n$ , set  $[\alpha]_n = [\alpha][\alpha + 1] \cdots [\alpha + n - 1]$ . Set  $[\alpha]_0 = 1$ .

**9.2. Lemma.** For  $\alpha_1, \alpha_4, \alpha_5, \dots, \alpha_{r+1} \in \mathbb{C}$  and

$$\zeta = \frac{r-3}{2}\alpha_1 + \frac{r-1}{2} - \alpha_4 - \cdots - \alpha_{r+1},$$



we have

$$\begin{aligned} & {}_{r+1}W_r(q^{2\alpha_1}; q^{2\alpha_4}, \dots, q^{2\alpha_{r+1}}; q^2, q^{2\zeta}) \\ &= \sum_{n \geq 0} \frac{[\alpha_1]_n}{[n]!} \cdot \frac{[\alpha_4]_n}{[1 + \alpha_1 - \alpha_4]_n} \cdot \dots \cdot \frac{[\alpha_{r+1}]_n}{[1 + \alpha_1 - \alpha_{r+1}]_n} \cdot \frac{[\alpha_1 + 2n]}{[\alpha_1]}. \end{aligned} \quad (9.2a)$$

**Proof.** The parameters  $a_2 = qa_1^{1/2}$ ,  $a_3 = -qa_1^{1/2}$ ,  $b_1 = a_1^{1/2}$ ,  $b_2 = -a_1^{1/2}$  contribute the following factor in (9.1c):

$$\begin{aligned} \frac{(qa_1^{1/2}; q)_n (-qa_1^{1/2}; q)_n}{(a_1^{1/2}; q)_n (-a_1^{1/2}; q)_n} &= \prod_{s=0}^{n-1} \frac{(1 - q^{s+1}a_1^{1/2})(1 + q^{s+1}a_1^{1/2})}{(1 - q^s a_1^{1/2})(1 + q^s a_1^{1/2})} \\ &= \prod_{s=0}^{n-1} \frac{1 - q^{2s+2}a_1}{1 - q^{2s}a_1} = \frac{1 - q^{2n}a_1}{1 - a_1} \end{aligned}$$

Since in the lemma we replace  $q$  by  $q^2$  and  $a_1$  by  $q^{2\alpha_1}$ , this factor is  $\frac{[\alpha_1 + 2n]}{[\alpha_1]} q^{2n}$ .

For any  $\alpha \in \mathbb{C}$ , we have

$$(q^{2\alpha}; q^2)_n = (q^{-1} - q)^n q^{n\alpha + \frac{n(n-1)}{2}} [\alpha]_n.$$

Substituting this formula in the definition of (9.1c) we obtain the claim of the lemma. The value of  $\zeta$  is chosen so that the powers of  $q$  cancel and do not appear on the right-hand side of (9.2a).

### 9.3. Identification of trigonometric $6j$ -symbols

An  ${}_{r+1}\phi_r$  series (9.1.a) is called balanced if  $z = q$  and

$$b_1 b_2 \cdots b_r = qa_1 a_2 \cdots a_{r+1}.$$

It was shown in [KR] that the  $6j$ -symbols without spectral parameter considered up to monomial factors are equal to balanced  ${}_4\phi_3$ , where the six parameters of  $6j$ -symbols are certain linear combinations of the logarithms of the six parameters of balanced  ${}_4\phi_3$ . It is also known [GR] that balanced  ${}_4\phi_3$  are related to very-well-poised  ${}_8\phi_7$  so that, again, logarithms of six parameters of both series are linear combinations of each other.

We will now give an analogous identification of trigonometric  $6j$ -symbols with certain basic hypergeometric series. Let  $a, b, c, A, B, C$  be nonnegative integers such that the triples  $(a, b, c)$ ,  $(A, b, C)$ ,  $(A, B, c)$ ,  $(a, B, C)$  are admissible. Set

$$\alpha = \frac{b+c-a}{2}, \quad \beta = \frac{a+c-b}{2}, \quad \gamma = \frac{a+b-c}{2}, \quad \sigma = \frac{a+b+c}{2}$$

and  $\mu = \max(A - \alpha, B - \beta, C - \gamma)$ . Replacing in Theorem 8.10 the numbers  $i, j, k, l, m, n$  by  $A, b, a, B, c, C$ , respectively, we obtain

$$\begin{aligned} & \left\{ \begin{matrix} A & b & C \\ a & B & c \end{matrix} \right\}' (x) \\ &= \sum_r (-1)^r \Delta(r) \left\{ \begin{matrix} A & \alpha & r \\ \beta & B & c \end{matrix} \right\} \left\{ \begin{matrix} B & \beta & r \\ \gamma & C & a \end{matrix} \right\} \left\{ \begin{matrix} C & \gamma & r \\ \alpha & A & b \end{matrix} \right\} \\ & \times (\theta(r, \beta, B) \theta(r, \gamma, C) \theta(r, \alpha, A))^{-1} \prod_{s=0}^{\sigma} \left[ x + \frac{\sigma - r}{2} - s - 1 \right] \left[ x + \frac{\sigma + r}{2} - s \right]. \end{aligned}$$

Denote by  $\lambda(A, B, C, a, b, c \mid x)$  the term on the right-hand side corresponding to  $r = \mu$ .

**9.4. Theorem.** *For any  $x \in \mathbb{C}$ ,*

$$\left\{ \begin{matrix} A & b & C \\ a & B & c \end{matrix} \right\}' (x) = \lambda(A, B, C, a, b, c \mid x) \times {}_{10}W_9(q^{2\alpha_1}; q^{2\alpha_4}, \dots, q^{2\alpha_{10}}; q^2, q^2),$$

where  ${}_{10}W_9$  is a balanced very-well-poised series with the following parameters:  $\alpha_1, \alpha_4, \alpha_5$  are the numbers

$$\frac{\mu + A - \alpha}{2} + 1, \frac{\mu + B - \beta}{2} + 1, \frac{\mu + C - \gamma}{2} + 1$$

so that  $\alpha_1$  is the maximal of them (and equals  $\mu + 1$ ), and  $\alpha_6, \dots, \alpha_{10}$  are the numbers

$$\frac{\mu - A - \alpha}{2}, \frac{\mu - B - \beta}{2}, \frac{\mu - C - \gamma}{2}, \frac{\mu + \sigma}{2} - x + 2, \frac{\mu + \sigma}{2} + x + 1.$$

**Proof.** Since  $c = \alpha + \beta$ , it follows from (6.2.c) and the tetrahedral symmetry for  $6j$ -symbols without spectral parameter that for any  $r = \mu + 2n$ ,

$$\left\{ \begin{matrix} A & \alpha & r \\ \beta & B & c \end{matrix} \right\} = \left\{ \begin{matrix} A & \alpha & \mu \\ \beta & B & c \end{matrix} \right\} [\mu + 1]_{2n} \frac{\left[ \frac{\mu + A - \alpha}{2} + 1 \right]_n}{\left[ \frac{\mu + B - \beta}{2} + 1 \right]_n}.$$

It follows from (6.1a) that

$$\theta(r, A, \alpha) = \theta(\mu, A, \alpha) [\mu + 1]_{2n} \frac{\left[ \frac{\mu + A + \alpha}{2} + 2 \right]_n \left[ \frac{\mu + A - \alpha}{2} + 1 \right]_n \left[ \frac{\mu + \alpha - A}{2} + 1 \right]_n}{\left[ \frac{\mu - \alpha - A}{2} \right]_n}.$$

Thus we obtain

$$\begin{aligned}
 & \frac{\left\{ \begin{matrix} A & \alpha & r \\ \beta & B & c \end{matrix} \right\} \left\{ \begin{matrix} B & \beta & r \\ \gamma & C & a \end{matrix} \right\} \left\{ \begin{matrix} C & \gamma & r \\ \alpha & A & b \end{matrix} \right\}}{\theta(r, A, \alpha)\theta(r, B, \beta)\theta(r, C, \gamma)} \\
 &= \frac{\left\{ \begin{matrix} A & \alpha & \mu \\ \beta & B & m \end{matrix} \right\} \left\{ \begin{matrix} B & \beta & \mu \\ \gamma & C & a \end{matrix} \right\} \left\{ \begin{matrix} C & \gamma & \mu \\ \alpha & A & b \end{matrix} \right\}}{\theta(\mu, A, \alpha)\theta(\mu, B, \beta)\theta(\mu, C, \gamma)} \\
 &\times \frac{[\frac{\mu+A-\alpha}{2} + 1]_n [\frac{\mu+B-\beta}{2} + 1]_n [\frac{\mu+C-\gamma}{2} + 1]_n}{[\frac{\mu+\alpha-A}{2} + 1]_n [\frac{\mu+\beta-B}{2} + 1]_n [\frac{\mu+\gamma-C}{2} + 1]_n} \\
 &\times \frac{[\frac{\mu-A-\alpha}{2}]_n [\frac{\mu-B-\beta}{2}]_n [\frac{\mu-C-\gamma}{2}]_n}{[\frac{\mu+A+\alpha}{2} + 2]_n [\frac{\mu+B+\beta}{2} + 2]_n [\frac{\mu+C+\gamma}{2} + 2]_n}.
 \end{aligned}$$

We also have

$$\Delta(r) = \Delta(\mu) \frac{[\mu + 2n + 1]}{[\mu + 1]}.$$

Next we transform the terms containing the parameter  $x$ :

$$\begin{aligned}
 \prod_{s=0}^{\sigma} [x + \frac{\sigma - \mu}{2} - n - s - 1] &= \prod_{s=0}^{\sigma} [x + \frac{\sigma - \mu}{2} - s - 1] \frac{[-x + \frac{\sigma + \mu}{2} + 2]_n}{[-x + \frac{\mu - \sigma}{2} + 1]_n}, \\
 \prod_{s=0}^{\sigma} [x + \frac{\sigma + \mu}{2} + n - s] &= \prod_{s=0}^{\sigma} [x + \frac{\sigma + \mu}{2} - s] \frac{[x + \frac{\sigma + \mu}{2} + 1]_n}{[x + \frac{\mu - \sigma}{2}]_n}.
 \end{aligned}$$

Now, we note that the range of summation in  $r$  is restricted from below by  $\mu$ , and that  $n = (r - \mu)/2$  varies over  $0, 1, 2, \dots$ . Thus we obtain the series  ${}_{10}W_9$  with the parameters as in the conditions of the theorem. Finally, it is straightforward to check that for these values of the parameters this series is balanced.

### 9.5. The Bailey transform and the Jackson summation

Numerous summation and transformation formulas are known for balanced and/or very-well-poised series. The transformation that contains most of the others as degenerate cases is the following Bailey formula for terminating bal-

anced very-well-poised series  ${}_{10}W_9$ ; see [GR]:

$${}_{10}W_9(q^{2\alpha_1}; q^{2\alpha_4}, \dots, q^{2\alpha_{10}}; q^2, q^2) = {}_{10}W_9(q^{2\beta_1}; q^{2\beta_4}, \dots, q^{2\beta_{10}}; q^2, q^2) \times \quad (9.5.a)$$

$$\times \frac{[\alpha_1 + 1]_n [\beta_1 + 1 - \alpha_7]_n [\beta_1 + 1 - \alpha_8]_n [\alpha_1 + 1 - \alpha_7 - \alpha_8]_n}{[\beta_1 + 1]_n [\alpha_1 + 1 - \alpha_7]_n [\alpha_1 + 1 - \alpha_8]_n [\beta_1 + 1 - \alpha_7 - \alpha_8]_n}$$

where  $n = -\alpha_{10}$  is a nonnegative integer,

$$\beta_1 = 2\alpha_1 + 1 - \alpha_4 - \alpha_5 - \alpha_6 = -\alpha_1 - 1 + \alpha_7 + \alpha_8 + \alpha_9 + \alpha_{10}, \quad (9.5b)$$

$$\beta_4 = \beta_1 - \alpha_1 - \alpha_4, \quad \beta_5 = \beta_1 - \alpha_1 + \alpha_5, \quad \beta_6 = \beta_1 - \alpha_1 + \alpha_6,$$

and  $\{\beta_7, \beta_8, \beta_9, \beta_{10}\}$  is an arbitrary permutation of  $\{\alpha_7, \alpha_8, \alpha_9, \alpha_{10}\}$ .

As a corollary of the identification of trigonometric  $6j$ -symbol with the balanced very-well-poised hypergeometric function, we obtain a geometric proof of the Bailey transformation formula. By Theorem 6.3.1,

$$\left\{ \begin{matrix} A & b & C \\ a & B & c \end{matrix} \right\} (x) = \left\{ \begin{matrix} a & B & C \\ A & b & c \end{matrix} \right\} (x). \quad (9.5c)$$

For concreteness, we assume that  $\mu = A - \alpha$  in Theorem 9.4, then

$$\alpha_1 = \mu + 1 = A + \frac{a - b - c}{2} + 1.$$

Set

$$\begin{aligned} \alpha_4 &= \frac{\mu + B - \beta}{2} + 1 = \frac{A + B - c}{2} + 1, \\ \alpha_5 &= \frac{\mu + C - \gamma}{2} + 1 = \frac{A + C - b}{2} + 1, \\ \alpha_6 &= \frac{\mu - C - \gamma}{2} = \frac{A - C - b}{2}, \\ \alpha_7 &= \frac{\mu - B - \beta}{2} = \frac{A - B - c}{2}, \\ \alpha_8 &= \frac{\mu - A - \alpha}{2} = \frac{a - b - c}{2}, \\ \alpha_9 &= \frac{\mu + \sigma}{2} - x + 2 = \frac{A + a}{2} - x + 2, \\ \alpha_{10} &= \frac{\mu + \sigma}{2} + x + 1 = \frac{A + a}{2} + x + 1. \end{aligned}$$

Then the parameters (9.5.b) in the Bailey transform have the following form:

$$\beta_1 = a + \frac{A - c - B}{2} + 1, \quad \beta_4 = \frac{a + b - c}{2} + 1,$$

$$\beta_5 = \frac{a + C - B}{2} + 1, \quad \beta_6 = \frac{a - B - C}{2},$$

$$\beta_7 = \frac{a - b - c}{2}, \quad \beta_8 = \frac{A - B - c}{2},$$

$$\beta_9 = \frac{A + a}{2} - x + 2, \quad \beta_{10} = \frac{A + a}{2} + x + 1.$$

Thus the change of the parameters in the Bailey transform corresponds precisely to the change of the parameters of  $6j$ -symbols under the transformation (9.5c). A direct calculation shows the coincidence of the corresponding monomials in both transforms. Finally, we note that since  ${}_{10}W_9$  is a rational function of variables  $\alpha_1, \alpha_4, \dots, \alpha_{10}$ , the identity (9.5.a), which is valid at infinitely many points corresponding to the integer parameters of the  $6j$ -symbols is also valid for any complex  $\alpha_1, \alpha_4, \dots, \alpha_{10}$  satisfying the only condition that at least one of them is a nonpositive integer. This condition guarantees that the series (9.2.a) is terminating.

The summation formula which contains most of known summation formulas as special cases is the Jackson formula for the balanced very-well-poised  ${}_8W_7$ , see [GR]:

$${}_8W_7(q^{2\alpha_1}; q^{2\alpha_4}, \dots, q^{2\alpha_8}; q^2, q^2) = \quad (9.5d)$$

$$\frac{[\alpha_1 + 1]_n [\alpha_1 + 1 - \alpha_4 - \alpha_5]_n [\alpha_1 + 1 - \alpha_4 - \alpha_6]_n [\alpha_1 + 1 - \alpha_5 - \alpha_6]_n}{[\alpha_1 + 1 - \alpha_4]_n [\alpha_1 + 1 - \alpha_5]_n [\alpha_1 + 1 - \alpha_6]_n [\alpha_1 + 1 - \alpha_4 - \alpha_5 - \alpha_6]_n}$$

where  $n = -\alpha_8$  is a nonnegative integer. The Jackson formula directly follows from the computation of the degenerate trigonometric  $6j$ -symbol given in Section 8.8. Indeed, set  $a = B + C$  in (9.5.c). Then

$$\alpha_4 = 1 + \alpha_1 - \alpha_5, \quad \alpha_5 = 1 + \alpha_1 - \alpha_4$$

so that four terms in the expression (9.2.a) for  ${}_{10}W_9$  cancel. The corresponding balanced very-well-poised  ${}_{10}W_9$  turns into a balanced very-well-poised  ${}_8W_7$ .

Combining (6.2.c) and (8.7.a) we obtain an expression for a degenerate trigonometric  $6j$ -symbol. Under the identification provided by Theorem 9.4, this expression becomes the Jackson summation formula.

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